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Index Provider Engagement and Climate Action: A Field Experiment*

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Abstract

Can rules-based engagement by an index provider affect corporate climate action? We present a preregistered field experiment with a major index provider whose climate indices are designed to exclude firms without a science-based climate target. The provider's CEO wrote to the board chairs of 300 firms, randomly drawn from 1,227 index constituents without such a target, reiterating the public exclusion rules. Receiving the letter raises the share of firms that commit to or set a climate target by 6.4 percentage points within two years. It also causes firms to reduce their reported carbon emissions by about 18% after four years. We do not detect an effect on firm value in either direction. Our results suggest that index provider engagement can spur corporate climate action by making existing exclusion rules salient.

Keywords: passive investing, index governance, climate targets, carbon emissions, field experiment

JEL classification: C93, D22, G23, G34, M14, Q54

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1 Introduction

Index providers have become increasingly powerful as passive funds and benchmark-linked mandates account for a growing share of global wealth. The rules governing index inclusion and weighting matter as they guide which firms passive capital is allocated to. Yet little is known about whether index providers can influence corporate behavior through the targeted communication of inclusion rules. We call this form of communication *index provider engagement* and examine it in the context of climate-focused indices. Specifically, we ask: can index provider engagement regarding climate-related rules affect corporate climate action?

We examine this question in a randomized, preregistered field experiment involving a sample of 1,227 listed firms, jointly accounting for roughly 39% of the global market capitalization of listed firms.[1] We collaborate with an index provider that offers a family of climate benchmark indices.[2] The provider's index rules required gradual exclusion of firms without climate targets verified by the Science Based Targets initiative (SBTi). Our sample comprises all firms in the provider's climate indices that do not yet have an eligible climate target and, following our preregistration, are not already targeted by Climate Action 100+, a large investor pressure campaign with similar goals.

Our treatment is a letter from the index provider's CEO to the board chair, stating that eligible climate targets are required for continued index inclusion and asking the firm to commit to setting such a target by the end of 2022. The index rules are publicly available; the letter therefore conveys no information that firms could not have obtained on their own. We randomly assign this treatment to 300 of the 1,227 eligible firms. We document three main results, summarized in Figure 1.

[1]The preregistration is available here.
[2]The provider offers Paris-Aligned and Climate Transition indices in line with European Commission Delegated Regulation (EU) 2020/1818.

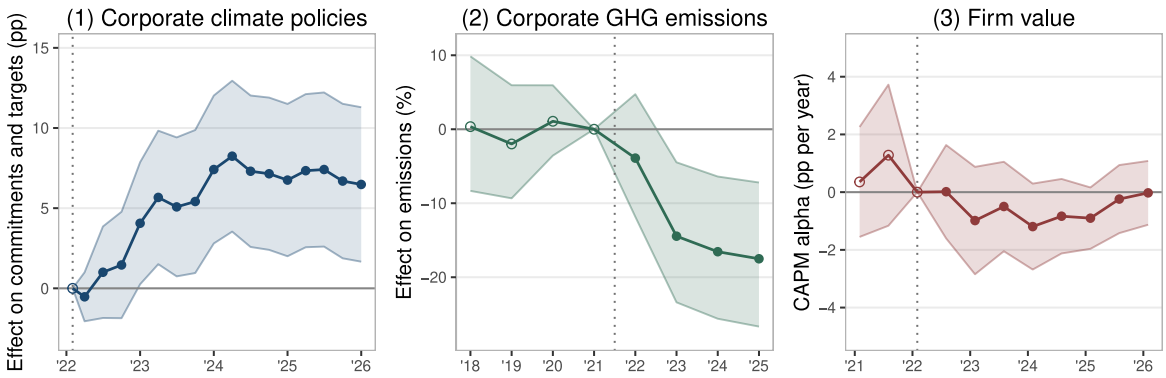


Figure 1. Main results. Estimated treatment effects of the letter (sent February 2022; dotted vertical lines). Panel (1): effect on the probability of an SBT commitment or set target at quarterly cutoffs from April 2022 through January 2026 (percentage points; linear probability models with controls). Panel (2): effect on carbon emissions (Scope 1 plus Scope 2) by reporting year (percent; Poisson pseudo-maximum likelihood (PPML) model with firm and year fixed effects). Panel (3): annualized CAPM alpha of the treated-minus-control portfolio over expanding windows from the letter at half-yearly horizons (percentage points per year). Shaded bands are 90% confidence intervals.

(1) Corporate climate policies. The treatment has a persistent effect on corporate climate policy. After two years, it raises the share of firms that have committed to or set a science-based target by 6.4 percentage points (28.0% of treated versus 21.6% of control firms; $p = 0.020$, randomization inference). The gap remains significant at the 10% level four years after the letter, having narrowed only slightly. The effect runs through commitments in the first year and through set targets by the second, in line with the SBTi's 24-month conversion deadline. The effect strengthens slightly with firm-level covariates, and is also significant for the higher-emitting half of the sample, which accounts for 98% of its pre-treatment carbon emissions.

(2) Reported emissions. The treatment also causes firms to reduce their reported greenhouse gas emissions. In a difference-in-differences comparison with firm and year fixed effects, the gap in annual direct emissions widens gradually to 18% by the fourth reporting year ($p = 0.002$, ran-

domization inference), with no significant pre-trend.[3] The decline is also significant among above-median emitters, appears to be driven largely by a subset of large emitters that cut their reported emissions by half or more, and is most prominent for firms' direct emissions. The reduction in reported emissions is robust to alternative estimators and sample choices, and it is also evident in an alternative emissions dataset sourced from the Carbon Disclosure Project (CDP). Reported cuts need not reflect physical abatement. CDP disclosures suggest that large treated emitters are more likely to rely on divestments, which could explain the sharp cuts among these firms.

(3) Firm value. The treatment has no detectable effect on firm value. The return on a treated-minus-control portfolio formed when the letters were sent is statistically indistinguishable from zero at every horizon out to four years, under alternative portfolio and factor-model choices, and for below- and above-median emitters alike (an annualized CAPM alpha of -0.02 percentage points per year, $p=0.984$, randomization inference). Over the full four-year window, the minimum detectable effect on the annualized long-short alpha is roughly 1.9 percentage points per year; smaller effects, plausible for an intervention of this size, would go undetected.

In sum, the experiment shows that index provider engagement has a substantial impact on firms' climate policy and outcomes. Holding index rules fixed, we show that the proactive communication of index rules – a form of voice – has an economically meaningful effect.

Frictions in managerial attention could explain these results. The relevant index rules were public before our treatment, yet firms respond strongly to a single letter from the index provider's CEO pointing to these rules. The letter conveys no information that firms could not have obtained; rather, it raises the salience of an existing rule and, more broadly, of the growing investor demand for credible corporate climate strategy that such rules embody. Under this interpretation, published exit rules and direct outreach are complements: an exit threat

[3]The fourth-year gap corresponds to an additional reduction of about 4.7% per year, against an estimated decline of about 1.0% per year among control firms. The resulting pace for treated firms is roughly what a 1.5°C pathway requires (IPCC, 2022).

embedded in the index methodology persuades firms to take action once it is made salient to the issuer. Section 6 develops this interpretation and discusses alternatives.

Contribution and related literature. We contribute to three literatures. First, we add causal evidence on the governance role of index providers. A growing literature documents that passive investing and index rules have important governance consequences (Appel, Gormley, and Keim, 2016; Hirst and Kastiel, 2019; Bebchuk and Hirst, 2019; Chattopadhyay, Shaffer, and Wang, 2020). We show that the salience of index rules matters, not only their existence: a letter restating the provider's published exclusion rules to board chairs raises climate commitments and lowers reported emissions, even when the rules stay fixed.

Second, we add evidence from a randomized experiment to the literature on the real effects of sustainable investing. Theory predicts that both exit (screens and capital reallocation) and voice can shift firm behavior (Pástor, Stambaugh, and Taylor, 2021; Edmans, Levit, and Schneemeier, 2022; Broccardo, Hart, and Zingales, 2022), yet the evidence on exit-driven real effects is mixed (Heath, Macciocchi, Michaely, and Ringgenberg, 2023; Gantchev, Giannetti, and Li, 2022; Berk and van Binsbergen, 2025; Eskildsen, Ibert, Jensen, and Pedersen, Forthcoming; Hartzmark and Shue, 2026). Studies of investor engagement generally find positive effects on corporate and environmental outcomes (Becht, Franks, Mayer, and Rossi, 2009; Dimson, Karakaş, and Li, 2015; Dyck, Lins, Roth, and Wagner, 2019; Barko, Cremers, and Renneboog, 2021; Hoepner, Oikonomou, Sautner, Starks, and Zhou, 2024; Naaraayanan, Sachdeva, and Sharma, Forthcoming; Becht, Franks, Miyajima, and Suzuki, 2023). However, as the sample of engaged firms is chosen rather than assigned, these estimates are potentially affected by selection. Indeed, coordinated engagements target large, visible firms in which the lead investors hold substantial stakes (Dimson, Karakaş, and Li, Forthcoming), climate engagements select on carbon footprint and transition-risk exposure (Derrien, Garel, Romec, and Zhou, 2025), and the effects of an investor coalition concentrate among firms selected on a discretionary basis (Hastreiter, 2025).

Engagement outcomes may also depend on the sender's own incentives (Lowry, Wang, and Wei, 2026), and outcomes are not easy to monitor for asset owners (Bauer, Cooper, van der Kroft, and Smeets, 2026). We randomize which firms receive outreach, thus ruling out selection bias in engagement by design. Because we hold the exit rule fixed while randomizing voice, the results support the view that exit and voice are complements: the published exclusion schedule prompted a substantially larger number of firms to act once a letter drew attention to it.

Third, we add to the literature on corporate climate commitments and their consequences for emissions. Observational and quasi-experimental studies link targets and pledges to subsequent reductions: Bolton and Kacperczyk (2023) document lower emissions growth among target-setters; Coqueret, Giroux, and Qiu (2026) use quasi-experimental variation in reporting infrastructure to show that SBTi adoption reduces emission intensity; and Bauer, Huber, Offner, and Wilms (2026) find that voluntary decarbonization pledges by U.S. firms are followed by price gains and lower emissions. Cenedese, Han, and Kacperczyk (2026) show that the threat of future exclusion from net-zero portfolios is likewise priced, with firms closer to exclusion trading at lower valuations. How reductions are achieved matters for interpreting them: recent results show that firms often lower reported emissions by divesting carbon-intensive assets rather than reducing emissions operationally (Berg, Ma, and Streitz, 2026; Duchin, Gao, and Xu, 2025). Our results show that both targets and reductions in reported emissions can be triggered by communicating specific exit threats.

Implications and external validity. Index rules and direct communication appear to be complements: published exclusion rules prompted action once a letter made them salient. Because the response we detect is in reported emissions, and part of it appears to run through divestments, rules that seek physical decarbonization may need to reward physical abatement rather than reductions in reported corporate emissions. Our setting is specific: a CEO-signed

letter and dedicated climate benchmarks with a published exclusion path. Whether similar effects arise for other types of index inclusion rules remains open for further research.

The remainder of the paper proceeds as follows. Section 2 describes the experimental design and data. Section 3 reports the effect on corporate climate policies, Section 4 the effect on reported emissions, and Section 5 the absence of a detectable effect on firm value. Section 6 discusses what might drive the firm response, and Section 7 concludes.

2 Experimental Design and Data

2.1 Setting

For our experiment, we worked together with an index provider that offers a family of climate benchmark indices. The provider supplied the list of constituent firms and administered the treatment as part of its regular operations; we, the authors, performed the randomization and observed the outcomes independently of the index provider.[4]

The provider's index rules required the gradual exclusion of firms without a climate target verified by the Science Based Targets initiative (SBTi). The climate benchmark indices, launched in 2020, comply with the European regulation on climate benchmarks[5] and followed a common exclusion schedule. Specifically, the index weights of firms without any commitment to set a science-based target (SBT) were gradually reduced, reaching zero from March 2025 onward; the weights of firms that had committed but did not yet have a verified SBT were likewise gradually reduced, with full exclusion scheduled for March 2030. The provider's most ambitious climate index moved both deadlines forward, to March 2023 and March 2025.

[4]The index provider has the option to remain anonymous. We additionally registered the trial in the AEA RCT Registry, as that registry allows the index provider's identity to be shared confidentially.

[5]Commission Delegated Regulation (EU) 2020/1818, which sets minimum standards for EU Climate Transition Benchmarks and EU Paris-aligned Benchmarks.

In addition, the indices overweighted firms that achieved a consistent reduction in annual greenhouse gas intensity of 7% or more. These rules and schedules were laid out in the provider's publicly available index methodology before the experiment. In December 2025, near the end of our observation window, the methodology was revised: the strict SBTi-based exclusions were replaced by an incremental underweighting of firms without a commitment or set target.

The SBTi is considered the gold standard for setting corporate climate goals. It encourages companies to join its initiative by committing to set SBTs and later submitting them for verification. As of August 2026, the SBTi's public registry contains more than 14,000 companies with either an SBT commitment or a set target. Because the SBTi is an NGO unrelated to the index provider, its registry also allows us to track firms' commitments and targets independently.

Setting an SBT follows a clear two-stage process. First, a company commits by sending a letter to the SBTi, pledging to develop a science-based target and submit it for validation within 24 months. Second, the SBTi validates the submitted target if it fulfills its criteria, which are designed to align corporate targets with the Paris Agreement's goal of limiting global warming to 1.5 degrees Celsius. Substantial and growing capital is linked to climate benchmarks, though the exact amount tracking them is uncertain. The index provider we work with is one of several offering EU-regulated climate benchmarks: ETFs tracking these indices globally held roughly USD 18 billion in assets just before the letters were sent (January 2022) and roughly USD 125 billion as of early 2026, according to Morningstar Direct data. The assets under management in ETFs directly tracking the index provider's climate indices alone amounted to roughly USD 45 million in early 2022. In addition, the provider's indices are licensed by financial institutions with combined assets under management of several trillion USD, including large asset owners who partnered to develop the indices. We do not have reliable information about what fraction of that capital is benchmarked against these specific indices, and companies in our sample likely faced the same uncertainty about potential trading flows from index reweighting.

2.2 Sample and randomization

Our sample comprises firms included in the provider's climate indices for which the treatment was relevant: namely, firms that had neither committed to nor set an SBT and were not on the Climate Action 100+ list. From the full constituent list of 1,957 unique firms,[6] we removed 669 firms with an existing SBTi commitment or target and 61 firms on the Climate Action 100+ list. Climate Action 100+ is an investor initiative that engages the 170 largest listed emitters and asks them to set climate targets, so we expected our letter to provide little additional encouragement for these firms to do so. This leaves a sample of 1,227 firms (300 treated, 927 control), which we use in all analyses.[7,8]

The sample is economically substantial. With a pre-treatment combined market capitalization of USD 30 trillion and annual direct (Scope 1) emissions of roughly 1.35 gigatons of CO_2e, it accounts for roughly 39% of the market capitalization and 12% of the Scope 1 emissions of listed companies globally.[9] However, the sample does not include some of the world's largest emitters; Climate Action 100+ firms are excluded by design. Appendix Table A1 benchmarks the sample's industry composition against the full climate-index universe: the sample spans all NACE sectors and broadly tracks the universe, with modestly fewer firms in manufacturing and utilities.

The treatment group was selected in a blocked random design. The index provider had the capacity to send letters to 300 companies. Exploiting this capacity constraint, we randomly drew 300 companies from the sample. We blocked the randomization by country, assigning

[6] Index constituents as of December 20, 2021.

[7] Sixteen of these firms (3 treated, 13 control) already had an SBTi commitment or target by the time the letters were sent on February 1, 2022, either because they committed after the sample was finalized on January 12, 2022, or because earlier commitments were not considered in the sample construction owing to reporting lags. We exclude these prior adopters from the policy regressions with firm-level controls.

[8] The preregistration foresaw excluding firms that merge, are acquired, split up, or go bankrupt. Compustat deletion records identify 94 sample firms deleted after the letter, 64 of them through acquisitions or mergers. We do not apply this exclusion in the main analysis, because deletion after the letter is a post-treatment outcome and conditioning on it could introduce selection; Appendix Table A3 shows that the main results hold when these firms are dropped.

[9] We approximate the universe of listed companies with the MSCI ACWI IMI. Sources: MSCI ACWI IMI Index factsheet, December 31, 2021 (archived version); MSCI Net-Zero Tracker, March 2022.

treatment with the same probability in every country, as national policies have been shown to strongly predict firms' likelihood of setting climate targets (Bolton and Kacperczyk, 2023). The treatment and control groups are well balanced across the pre-treatment covariates reported in Table 1 and across countries and sectors (Appendix Tables A4 and A5).

Because treatment was assigned randomly, the design supports inference based on the randomization itself. Throughout the paper we therefore complement conventional standard errors with randomization inference, testing the sharp null of no treatment effect for any firm: treatment is re-assigned 2,000 times at the firm level within the country blocks of the original randomization, holding each country's number of treated firms fixed, and the statistic is re-computed on every draw. The reported p-value is the share of draws, counting the observed assignment itself, with an estimate at least as large in absolute value as the observed one. For the adoption shares, this is a non-parametric test of independence on the contingency table of treatment and outcome status, the test specified in the preregistration.

2.3 Treatment

Our treatment is a letter, signed by the index provider's CEO and addressed to the chairperson of each treated firm's board, requesting that the firm commit to setting an SBT through the SBTi. The letters were sent on February 1, 2022.

The letter spells out the consequences of not adopting an SBT for the firm's index inclusion. It starts by saying that the company is currently a constituent of at least one of the index provider's climate indices. It then points out that an SBT, verified by the SBTi, is required to remain a constituent of these climate indices in the future. It summarizes the relevant rules and schedules for continued index inclusion, as described in Section 2.1. The letter notes that EU regulations require providers of climate benchmarks to increase the weights of companies with evidence-based climate targets, and that the index provider goes further by requiring a target

Table 1. Pre-treatment balance

Means and standard deviations of pre-treatment covariates by treatment arm; the last column reports the normalized difference. Size is the logarithm of market capitalization in December 2021, in USD billions. Carbon emissions (Scope 1 plus Scope 2, location-based) and emission intensity (emissions per million USD of revenue) refer to fiscal year 2020, the latest fiscal year reported before the letter. The pre-treatment return is the cumulative return over February 2021 to January 2022. The ESG risk score is Sustainalytics' ESG Risk Rating as of December 2021 (higher values indicate higher unmanaged risk). Profitability (return on assets), book leverage, and book-to-market are from each firm's latest annual report ending on or before December 2021. A joint orthogonality test (treatment assignment regressed on all covariates with country fixed effects) yields $F = 0.67$, $p = 0.72$ ($N = 1{,}199$).

	Control		Treatment		
	Mean	SD	Mean	SD	Norm. diff.
Size (log market cap)	2.52	1.06	2.53	1.08	0.00
Carbon emissions (kt CO_2e)	839	1,903	848	1,781	0.00
Emission intensity (t CO_2e / USD m rev.)	115	221	128	243	0.06
Pre-treatment 12-month return	0.09	0.30	0.09	0.30	0.01
ESG risk score (Sustainalytics)	22.7	7.4	22.8	7.5	0.02
Profitability (ROA)	0.05	0.08	0.05	0.07	−0.05
Book leverage	0.29	0.22	0.27	0.19	−0.10
Book-to-market	0.54	0.57	0.56	0.55	0.03
Observations (firms)	927		300		

verified by the SBTi. It notes that the most ambitious index is backed by major asset owners and managers. The letter concludes with a direct request: it encourages the company to commit to setting an SBT by the end of 2022, asks it to fill out a brief survey explaining its current status and perspectives, and offers a dialogue with the index provider's sustainable investing experts.[10] The underlying index rules were in place and publicly available well before the letter was sent.

2.4 Outcomes and data

We estimate treatment effects on three outcomes: corporate climate policies, reported emissions, and firm value. The first is our preregistered primary outcome; the second comprises the

[10] Appendix Table A6 compares the subsequent SBTi outcomes of firms responding to the survey with the full treated sample. Replies from later adopters typically indicated that steps toward an SBTi submission were initiated; replies from non-adopters cited capacity constraints or a preference for the firm's own targets without SBTi validation, or requested that such matters not be addressed to the board chair.

preregistered secondary outcomes, carbon emissions and emission intensity; the firm-value analysis was not preregistered.

Corporate climate policies (SBTi targets). We measure corporate climate policies based on whether a firm has committed to or set an SBT, and track the two stages separately. The data on commitments and targets come from the SBTi's public registry, the same source used in, for example, Bolton and Kacperczyk (2023).[11] Every commitment and every validated target in the registry carries its own publication date: a firm counts as committed from the date it files its commitment letter and as having set a target from the date the SBTi validates it. We track commitments and targets at quarterly cutoffs from April 1, 2022 through January 1, 2026 and report results for January 1 of each year from 2023 to 2026 in the tables. In 2023, the SBTi began marking commitments that do not produce a validated target within 24 months as "removed" in its registry. Our main outcome is absorbing and does not reflect these expirations: we track whether a firm has ever committed to or set an SBT after the treatment. Appendix Figure A1 reports the results under an alternative outcome definition in which expired commitments are excluded. We match SBTi registry entities to sample firms by ISIN and, where no ISIN is available in the SBTi registry, by company name.[12]

Reported emissions (Trucost). Our main emissions outcome, a firm's *carbon emissions*, is the sum of its Scope 1 and Scope 2 (location-based, purchased-energy) greenhouse-gas emissions in tonnes of CO_2e, from S&P Trucost. For Scope 1 we use Trucost's "GHG Direct"

[11] We use the SBTi's target-dashboard records, retrieved in August 2026. The results also hold when the outcome is instead constructed from archived downloads of the registry (Appendix Table A7).

[12] We first accept unique exact ISIN matches in the August 2026 SBTi companies export, which link 336 sample firms. For the remaining firms, we transliterate and lower-case company names and remove punctuation, ISO 20275 legal forms, and the terms "group," "holding(s)," "public/publ," "and," and "the." We review a pair when the normalized names or their token sets are identical, when one name contains at least two tokens of the other, or when the names are identical once spaces are removed. Country and industry guide the review but do not determine the match. We accept a pair only when alternative identifiers or corporate records confirm that both refer to the same direct issuer; we do not match firms to their parents or subsidiaries. This procedure links 57 additional firms by name.

measure, covering all greenhouse gases from sources the firm owns or controls.[13] Firms report a fiscal year's emissions in the following calendar year; we therefore index reported emissions by reporting year, where reporting year k covers fiscal year $k-1$. We link the Trucost records to the sample by GVKEY. The emissions panel covers reporting years 2018 to 2025.[14] Where a firm's emissions series has an interior gap, we fill it by carrying the last reported value forward.[15] We winsorize emissions at the 95th percentile within each reporting year.[16]

CDP disclosures. As an alternative data source for reported carbon emissions and to examine the reasons firms give for their emission reductions, we draw on firms' own CDP climate disclosures. CDP, formerly the Carbon Disclosure Project, surveys firms on their climate data through an annual questionnaire on behalf of investors; it records each firm's self-reported Scope 1 and Scope 2 emissions and the reasons the firm gives for year-over-year changes in its emissions, each recorded as a percentage of the prior-year level together with its direction. Following Berg, Ma, and Streitz (2026), we harmonize CDP's reason categories across survey years into seven groups: divestments, boundary or methodology changes, acquisitions, changes in output, emission-reduction activities, physical operating conditions, and other or unidentified reasons. For each group, we construct a firm-level indicator equal to one if the firm cites that group as a reason for an emissions reduction within a given period. Not all firms disclose reasons for changes in their emissions; the sample for this analysis comprises the firms that report at least one reason within the analysis period. Like the Trucost series,

[13] Results are similar with Trucost's narrower "GHG Scope 1" measure, which includes only the greenhouse gases covered by the Kyoto Protocol (Appendix Table A22).

[14] Panels that add back the earlier first reported year or drop one more pre-treatment year give qualitatively unchanged results (Appendix Table A9).

[15] Our results are essentially unchanged if we instead omit firms with interior gaps from the balanced panel (Appendix Table A8).

[16] Estimates without winsorization or with caps at the 99th or 90th percentile are qualitatively unchanged (Appendix Table A10). Without winsorization, the year-specific estimates are similar but less precise. The corresponding randomization-inference p-values for reporting years 2023 to 2025 range from 0.012 to 0.023 (Appendix Table A11).

CDP-reported emission levels are winsorized at the 95th percentile within each reporting year. We match the CDP records to our sample by ISIN.

Stock returns. We measure firm value with monthly USD total returns built from Compustat (North America and Global) security files, using each firm's primary home-market listing and converting to USD at daily exchange rates. Returns are winsorized at the 1st and 99th percentiles within each month. We obtain monthly developed-market factor returns from Ken French's data library.

Firm covariates. Country and NACE sector are provided by the index provider. Pre-treatment market capitalization is the firm's Compustat market capitalization at the end of December 2021, in USD billions, and size is its logarithm. Thirty firms (25 control, 5 treated) lack at least one covariate and drop out of the covariate-adjusted specifications.[17] Pre-treatment carbon emissions and emission intensity are the fiscal-year 2020 Trucost values, winsorized at the 95th percentile. The ESG risk score is Sustainalytics' ESG Risk Rating; higher scores indicate more unmanaged ESG risk. We construct the accounting controls using Compustat annual fundamentals for each firm's latest fiscal year ending on or before December 2021: profitability (income before extraordinary items over total assets), book leverage (long-term debt plus debt in current liabilities, over total assets), and book-to-market (book equity over pre-treatment market capitalization). For cross-listed firms reported in both Compustat files, we keep the consolidated record with the largest total assets. The pre-treatment return is the firm's cumulative return over February 2021 to January 2022. The accounting controls and the pre-treatment return are winsorized at the 1st and 99th percentiles.

[17] Of these, 3 have no Compustat market capitalization, 8 no pre-treatment return, 12 no fiscal-year 2020 Trucost value, 7 no ESG risk score, 6 lack an accounting control, and 2 have no valid NACE sector.

Board networks (BoardEx). To test for spillovers between the treatment arms, we retrieve data on board overlaps from BoardEx. We flag control firms that share a board chairperson with a treated firm, the most direct channel through which the letter, addressed to the chair, could reach an untreated board. We match the BoardEx records to the sample by GVKEY, using the WRDS BoardEx–Compustat link file, and by ISIN for firms this file does not cover; 7 remaining firms are matched by name.

3 The effect on corporate climate policies

The treatment causes an increase in science-based-target adoption (Figure 2). At the first annual cutoff, January 1, 2023, the share of firms that have committed to or have set a science-based target is 3.0 percentage points higher in the treatment group (15.3% versus 12.3%), a difference that is not statistically significant ($p = 0.203$, randomization inference). At the January 2024 cutoff, two years after the letter, the effect is 6.4 percentage points (28.0% versus 21.6%; $p = 0.020$, randomization inference). The gap peaks on April 1, 2024, at 7.5 percentage points ($p = 0.007$, randomization inference). By January 2026 it is slightly smaller, at 5.9 percentage points, but remains significant at the 10% level ($p = 0.059$, randomization inference). The estimates become slightly larger and more precise when adding firm-level controls (Table 2).

The evolution of the effect over time aligns with the SBTi's timelines. Figure 2 splits the climate-policy outcome into its two stages: filed commitments not yet converted and set targets. The treatment raises commitments over the first twelve months, by 3.5 percentage points one year after the letter. By January 2024 the commitment gap recedes (+2.0 points) while the target-set gap opens (+4.4 points), in line with the SBTi's 24-month deadline for converting commitments into validated targets.

We also find a significant effect among the firms that matter most for aggregate emissions. Splitting the sample at the median of pre-treatment carbon emissions, the effect as of January

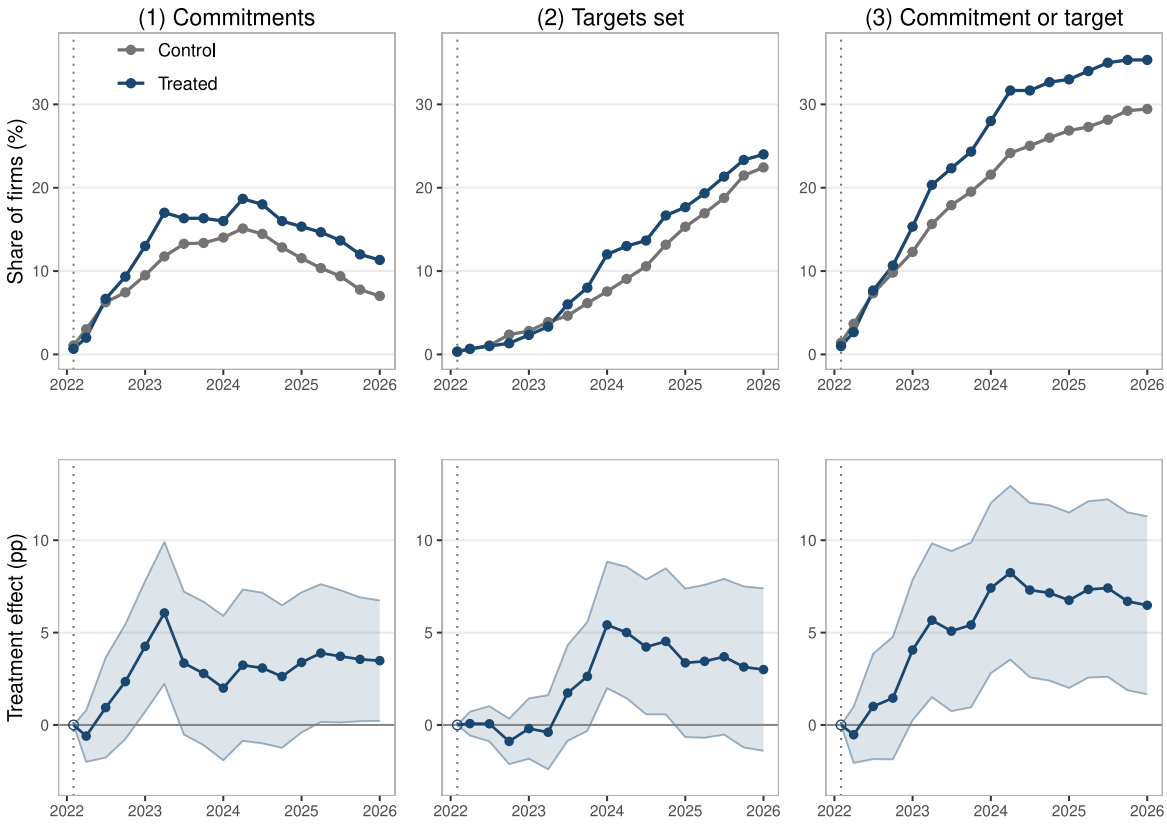


Figure 2. Treatment effect on corporate climate policies. Top row: shares of firms by treatment arm that have committed to an SBT but have not yet set a target (panel 1), have set an SBT (panel 2), and either (panel 3, the preregistered outcome), at the letter date and at quarterly cutoffs from April 1, 2022 through January 1, 2026. A firm counts as committed until it sets a target, even if the SBTi has removed its commitment in the meantime (Appendix Figure A1 shows point-in-time statuses). Bottom row: treatment effects on each probability from per-cutoff linear probability models with the full control set, one observation per firm; the letter-date point is zero by construction. Panels 1 and 2 are disjoint and together make up panel 3, so their effects sum to the panel 3 effect. Filled markers are post-treatment estimates; hollow markers are the letter-date reference; and shaded bands are 90% heteroskedasticity-robust confidence intervals. Appendix Table A2 reports the panel 3 estimates without and with controls, with standard errors and randomization-inference p-values.

2024 is significant and somewhat stronger among above-median emitters, which account for 98% of the sample's pre-treatment emissions (Table 3). However, the difference in effect size between the two halves is not significant (Appendix Table A12). The effect is likewise significant among above-median firms when splitting by pre-treatment emission intensity (Appendix Table A13).

Table 2. Treatment effect on corporate climate policies

Linear probability models for an SBT commitment or set target at annual cutoffs. Odd columns include treatment only; even columns add the pre-treatment covariates and country and NACE-sector fixed effects and exclude the 16 prior adopters. Coefficients are changes in the adoption probability. Heteroskedasticity-robust standard errors in parentheses; * $p<0.10$, ** $p<0.05$, *** $p<0.01$. RI p-values are from randomization inference as described in Section 2.2.

As of	Jan 2023		Jan 2024		Jan 2025		Jan 2026	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Treatment	0.030	0.041*	0.064**	0.074***	0.061**	0.068**	0.059*	0.065**
	(0.023)	(0.023)	(0.029)	(0.028)	(0.031)	(0.029)	(0.031)	(0.029)
Size		0.031***		0.047***		0.057***		0.056***
		(0.012)		(0.014)		(0.015)		(0.015)
ESG risk score		−0.004***		−0.008***		−0.009***		−0.011***
		(0.001)		(0.002)		(0.002)		(0.002)
Carbon emissions		0.018		0.015		0.004		0.004
		(0.013)		(0.015)		(0.015)		(0.016)
Emission intensity		−0.080		−0.164		−0.122		−0.095
		(0.104)		(0.118)		(0.121)		(0.128)
Pre-treatment return		0.032		0.086*		0.055		0.058
		(0.042)		(0.050)		(0.051)		(0.052)
Profitability (ROA)		−0.150		−0.400**		−0.171		−0.245
		(0.123)		(0.164)		(0.176)		(0.186)
Book leverage		0.045		0.070		0.066		0.043
		(0.042)		(0.052)		(0.055)		(0.056)
Book-to-market		−0.035*		−0.032		−0.045		−0.058**
		(0.020)		(0.027)		(0.028)		(0.029)
Country FE	No	Yes	No	Yes	No	Yes	No	Yes
NACE sector FE	No	Yes	No	Yes	No	Yes	No	Yes
RI p-value	0.203	0.050	0.020	0.005	0.038	0.016	0.059	0.026
Observations	1,227	1,181	1,227	1,181	1,227	1,181	1,227	1,181
R^2	0.002	0.089	0.004	0.141	0.003	0.163	0.003	0.183

In line with this pattern, weighting firms by their pre-treatment carbon emissions raises the January 2024 estimate to 14.2 percentage points ($p=0.033$, randomization inference; Appendix Table A14). Further, the estimate is unlikely to be contaminated by spillovers through board networks: control firms that share a board chair with a treated firm are no more likely to adopt commitments or targets at any point (Appendix Tables A15 and A16). If spillovers to control firms were nonetheless present, they would raise adoption in the control group, and our estimates would be a lower bound of the true effect.

Commitments in the treatment group are slightly more likely to expire, but the difference is not significant. When firms do not follow up with a validated target within 24 months, the

Table 3. Treatment effect on corporate climate policies by pre-treatment emissions

Linear probability models for an SBT commitment or set target as of January 2024, estimated separately for firms below and above the sample median of fiscal-year 2020 carbon emissions, without and with the pre-treatment covariates and country and NACE-sector fixed effects. Heteroskedasticity-robust standard errors in parentheses; * $p<0.10$, ** $p<0.05$, *** $p<0.01$. RI p-values are from randomization inference as described in Section 2.2.

As of	Jan 2024			
FY2020 emissions	Below median		Above median	
	(1)	(2)	(3)	(4)
Treatment	0.040	0.037	0.087**	0.105**
	(0.040)	(0.038)	(0.042)	(0.042)
Firm-level controls	No	Yes	No	Yes
Country FE	No	Yes	No	Yes
NACE sector FE	No	Yes	No	Yes
RI p-value	0.338	0.334	0.040	0.010
Observations	608	592	607	589
R^2	0.002	0.200	0.008	0.197

SBTi may mark their commitments as expired. Among the 223 sample commitments that were at least two years old as of January 2026, 25.0% of treated and 18.1% of control commitments had expired ($p=0.26$; Appendix Table A17). Appendix Figure A1 shows Figure 2 with expired commitments excluded: the gap in the combined outcome is mostly unaffected through 2024, but shrinks thereafter and is no longer significant by January 2026 (Appendix Table A18).

4 The effect on reported emissions

The treatment causes firms to reduce their reported carbon emissions. We estimate the effect year by year,

$$Emissions_{it} = \exp\left(\sum_{k\neq 2021} \beta_k \, Treat_i \times \mathbb{1}\{t=k\} + \mu_i + \theta_t\right)\eta_{it}, \tag{1}$$

where $Emissions_{it}$ are the carbon emissions firm i reports in year t, μ_i and θ_t are firm and year fixed effects, and η_{it} is a multiplicative error term with $\mathbb{E}[\eta_{it} \mid Treat_i, \mu_i, \theta_t] = 1$. Reporting year 2021 (fiscal year 2020), the last year reported before the letter, is the omitted reference, so each

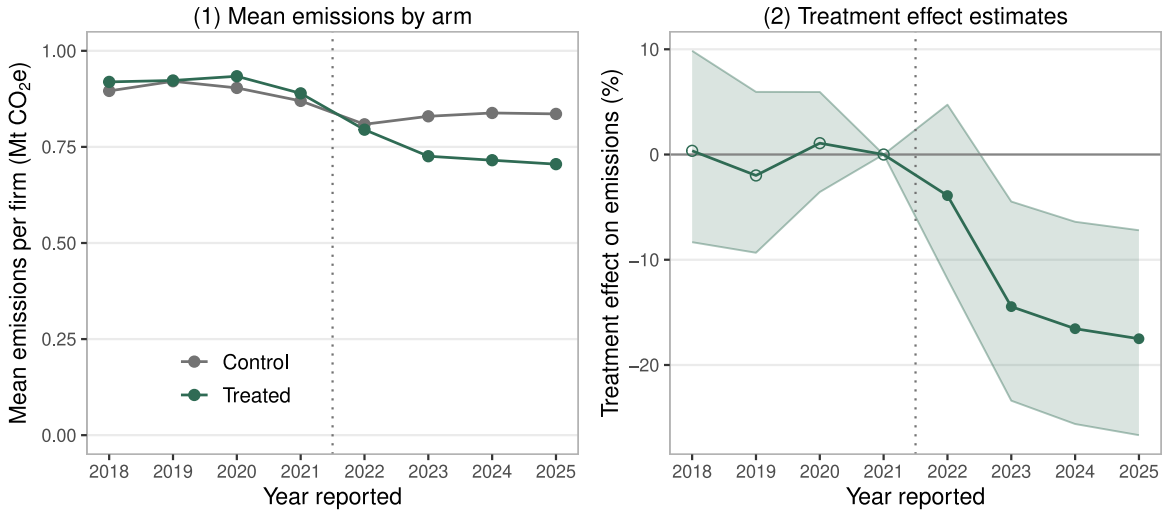


Figure 3. Reported carbon emissions: arm means and treatment effects. Panel (1): mean carbon emissions per firm by treatment arm and reporting year. Panel (2): treatment-by-year effects from equation (1), expressed as implied percent changes relative to reporting year 2021 ($100(e^{\beta}-1)$), with 90% confidence intervals from standard errors clustered by firm. Dotted lines mark the letter date. The coefficients are tabulated in Table 4.

β_k measures the treated–control gap in reporting year k relative to that reference year.[18] We estimate equation (1) by Poisson pseudo-maximum likelihood (PPML)[19] and cluster standard errors by firm; the estimation sample is the balanced panel of 1,120 firms with emissions data in every reporting year from 2018 to 2025. The results are qualitatively similar under alternative estimators: ordinary least squares on emissions, on log emissions, and on emission intensity (Appendix Table A20; Appendix Figure A2).[20]

[18] Firms report a fiscal year's emissions with a lag, typically late in the following calendar year. When the letters were sent in February 2022, fiscal-year 2021 emissions were thus not yet reported, and we cannot rule out that the treatment affected them; we therefore index emissions by reporting year throughout (Section 2.4). Classifying reporting year 2022 as pre-treatment leaves the pooled estimate similar and significant (Appendix Table A19).

[19] PPML is well-suited for this estimation as greenhouse-gas emissions are non-negative and highly right-skewed (Santos Silva and Tenreyro, 2006; Chen and Roth, 2024; Cohn, Liu, and Wardlaw, 2022).

[20] The log-OLS pre-treatment coefficients in Appendix Figure A2 are slightly positive but not significant, individually or jointly (Wald test, $p=0.76$). Among above-median emitters they are close to zero (Appendix Figure A4), so the pattern likely reflects lower pre-treatment data quality for small emitters.

Table 4. Treatment effect on reported carbon emissions, by year

Estimates of equation (1) on the balanced panel of 1,120 firms with emissions data in every reporting year from 2018 to 2025. The model is estimated by PPML with firm and year fixed effects, so the coefficients are semi-elasticities. Event time is in reporting years relative to the letter; the omitted reference $t-1$ is reporting year 2021, the last reported before the letter. A joint Wald test of the pre-treatment coefficients yields $p=0.64$. Observations: 8,960; within pseudo-R^2: 0.050. Standard errors clustered by firm in parentheses; * $p<0.10$, ** $p<0.05$, *** $p<0.01$. RI p-values are from randomization inference as described in Section 2.2.

Dependent variable:	Carbon emissions						
Event time:	$t-4$	$t-3$	$t-2$	$t+0$	$t+1$	$t+2$	$t+3$
Reporting year:	2018	2019	2020	2022	2023	2024	2025
Treatment × year	0.004	−0.020	0.011	−0.040	−0.156**	−0.181***	−0.192***
	(0.055)	(0.047)	(0.029)	(0.052)	(0.067)	(0.070)	(0.072)
RI p-value	0.954	0.680	0.743	0.383	0.002	0.001	0.002

The treatment effect on reported emissions is statistically significant from the second post-treatment reporting year (2023; $p=0.002$, randomization inference) and grows to about 18% by the fourth (Table 4 and Figure 3). The pre-treatment coefficients for reporting years 2018 through 2020 are small and statistically indistinguishable from zero, both individually and jointly (Wald test, $p=0.64$). The same pattern is visible in panel (1) of Figure 3, which plots unadjusted mean emissions by treatment arm: the two groups track each other closely through the pre-treatment years and diverge after the letter. Because PPML weights firms by their emission levels, the estimated treatment effects speak to the effect on aggregate emissions. Ordinary least squares on log emissions gives each firm equal weight and yields similar estimates, suggesting a comparable effect on the average firm (Appendix Table A20). The fourth-year estimate corresponds to an additional reduction of about 4.7% per year, against an estimated decline of about 1.0% per year among control firms. For the subgroup and robustness analyses that follow, we pool the post-treatment years,

$$Emissions_{it} = \exp\left(\beta\, Treat_i \times Post_t + \mu_i + \theta_t\right)\eta_{it}, \tag{2}$$

where $Post_t$ indicates reporting years 2022 onward and β is the average treatment effect across the post-treatment years, estimated by PPML with standard errors clustered by firm as before. The pooled estimate is -0.140, a reduction of about 13% ($p=0.005$, randomization inference; Table 5, column 1).

The treatment effect holds among the firms that matter most for aggregate emissions. Splitting the sample at the median of pre-treatment emissions, the effect among above-median emitters is statistically significant ($p=0.004$, randomization inference) and very close to the full-sample estimate (Table 5); for below-median firms, the point estimate is somewhat smaller and not statistically significant ($p=0.254$, randomization inference; Appendix Figures A3 and A4 show the corresponding year-specific estimates). The difference in effect size between the two halves is not significant ($p=0.690$, randomization inference; Appendix Table A21).

Consistent with this, the treatment effect on aggregate emissions appears to be largely driven by a subset of large emitters that strongly reduce their reported emissions. Table 6 sorts firms by the change in their reported emissions over the four post-treatment years into increasers, moderate reducers (cutting less than half), and heavy reducers (cutting half or more). The treatment raises the share of heavy reducers by 6.2 percentage points ($p=0.017$, randomization inference); we find no significant effect on the shares of increasers or moderate reducers. The shift toward heavy reducers is concentrated among large firms: heavy reducers account for 24.5% of the treated arm's pre-treatment emissions but only 7.5% of the control arm's pre-treatment emissions ($p=0.001$, randomization inference), with a comparably lower share of pre-treatment emissions held by moderate reducers ($p=0.004$, randomization inference). Appendix Figure A5 compares the two arms' distributions of emission changes, as shares of firms and of pre-treatment emissions.

The treatment effect is concentrated in firms' direct emissions. Estimating the effect separately for each scope, we find a reduction of about 18% in direct (Scope 1) emissions, which account for 75% of the pre-treatment emissions in our main emissions measure (Appendix Table A22). For purchased-energy (Scope 2) emissions, the remaining 25%, the point estimate

Table 5. Treatment effect on reported carbon emissions, by pre-treatment emissions

PPML estimates of equation (2) over reporting years 2018 to 2025; coefficients are semi-elasticities. Columns (2) and (3) split firms at the balanced-panel median of fiscal-year 2020 emissions. The post period comprises reporting years 2022 onward. Standard errors clustered by firm in parentheses; * $p<0.10$, ** $p<0.05$, *** $p<0.01$. RI p-values are from randomization inference as described in Section 2.2.

Dependent variable:	Carbon emissions		
	All firms	Below median	Above median
	(1)	(2)	(3)
Treatment × Post	−0.140**	−0.102	−0.141**
	(0.061)	(0.084)	(0.063)
Firm FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
RI p-value	0.005	0.254	0.004
Observations	8,960	4,480	4,480
Within pseudo R^2	0.010	0.002	0.010

Table 6. Treatment effect on the distribution of emission changes

Treated and control firms, sorted by their four-year post-treatment change in reported emissions (reporting years 2021 to 2025): increasers (no reduction), moderate reducers (less than 50%), and heavy reducers (50% or more). For each group, the share of the arm's firms and the share of the arm's reporting-year 2021 emissions, in percent. Stars are based on randomization-inference p-values from 2,000 re-assignments of treatment within country blocks: * $p<0.10$, ** $p<0.05$, *** $p<0.01$.

	Share of firms (%)			Share of pre-treatment emissions (%)		
	Treated	Control	Difference	Treated	Control	Difference
Increasers (no reduction)	36.5	42.1	−5.6	23.9	20.8	+3.1
Moderate reducers (<50%)	42.2	42.8	−0.6	51.5	71.7	−20.2***
Heavy reducers (≥50%)	21.3	15.0	+6.2**	24.5	7.5	+17.1***

is also negative but not statistically significant. We likewise find no effect on upstream Scope 3 emissions, which are not part of our main emissions measure because their measurement quality and comparability across firms are known to be problematic (Busch, Johnson, and Pioch, 2022; Klaaßen and Stoll, 2021). With this caveat in mind, the results do not suggest that reductions in direct emissions coincide with a shift into indirect emissions.

Robustness. We show that neither Trucost's estimation of undisclosed emissions, nor attrition, nor any individual firm is likely to drive our results.

Trucost's estimation of undisclosed emissions is unlikely to drive our results. Trucost estimates emissions for firms that do not fully disclose them. Among the 403 firms that disclosed 100% of their Scope 1 emissions in fiscal year 2020, the point estimate is -0.165, slightly larger in magnitude than the main estimate ($p=0.010$, randomization inference; conventional $p=0.08$; Appendix Table A23). Treated firms do not change how much they disclose: the difference-in-differences estimate of the treatment effect on the Trucost disclosure score itself is small and insignificant (Appendix Table A24). Finally, the effect also holds in the emissions that firms themselves report to CDP: -0.147 on all CDP-matched firm-years ($p=0.021$, randomization inference) and -0.206 on the balanced CDP panel ($p=0.067$, randomization inference; Appendix Table A25).

Attrition is unlikely to drive our results either. Our analysis uses the balanced panel of firms with emissions data in every reporting year from 2018 to 2025[21]; of the 1,226 firms with Trucost data (after filling interior gaps), this drops 47 without complete pre-treatment data and 59 that stopped reporting after the treatment. The share of firms that report throughout the post-treatment period is somewhat higher in the treatment group (97.2% versus 94.3%; $p=0.043$, randomization inference), suggesting that the treatment may have motivated firms to keep reporting. To bound the potential effect of this differential attrition, we retain every firm with complete pre-treatment data and impute any missing post-treatment years using two extreme scenarios. In the adversarial scenario, imputed control firms decarbonize faster than 95% of the observed control firms, while imputed treated firms decarbonize slower than 95% of the observed treated firms; the favorable scenario reverses both. Under both scenarios the coefficient remains close to the main estimate and statistically significant (Appendix Table A27).

[21] Our results also hold on the unbalanced panel including all firms with Trucost data (Appendix Table A26).

No individual firm drives our results. Dropping one firm at a time, the pooled coefficient ranges from -0.155 to -0.112 with a median of -0.140; all pooled estimates are significant at the 5% level. For the fourth-year estimate, the coefficient ranges from -0.205 to -0.165 and all estimates stay significant at the 5% level (Appendix Tables A28 and A29).

Reported versus physical reductions. Reductions in reported emissions need not necessarily imply physical decarbonization. A treated firm can lower its reported carbon emissions without abating in the real economy; specifically, it can redraw inventory boundaries or change accounting methods, or divest from carbon-intensive assets that remain in operation under new ownership (Berg, Ma, and Streitz, 2026; Duchin, Gao, and Xu, 2025). The timing of the observed effect on reported emissions is consistent with reporting choices playing a role. Reporting year 2022 covers fiscal year 2021, which was largely over when the letter arrived; notably, the estimate for that year is already negative (-4%), although not statistically significant ($p=0.38$, randomization inference; Table 4). To the extent that this early estimate reflects a response to the letter, it would have to come from choices made when the inventory was compiled rather than from changes in operations. To distinguish between the two, we draw on firms' CDP disclosures, which record reasons for year-over-year changes in emissions. We then test whether treated firms are more likely than control firms to attribute their emission reductions to accounting-based reasons.

The stated reasons suggest that large treated emitters are more likely to use divestment to reduce their reported emissions. Table 7 reports the share of firms by arm citing a given CDP reason category for a post-treatment emissions reduction, unweighted and weighted by pre-treatment emissions (Section 2.4). Reductions attributed to emission-reduction activities are nearly universal (92.5% of treated versus 92.7% of control firms), and no unweighted difference is statistically significant. Weighted by pre-treatment emissions, divestment-related reductions are more common among treated firms (43.5% versus 21.1% of pre-treatment emissions; $p=0.084$,

randomization inference). This finding should be treated with some caution, as only a limited number of firms state reasons for their emission changes (504 enter the comparison).

Table 7. Reported reasons for emission reductions

The share of firms by arm citing a given CDP reason category for a post-treatment emissions reduction, over reporting years 2022–2025. The first three columns are unweighted shares of firms; the last three weight each firm by its fiscal-year 2020 CDP-reported emissions. The sample comprises the 504 firms (120 treated) that attribute at least one emissions change to a reason during the period and report a fiscal-year 2020 CDP emissions level. Stars are based on randomization-inference p-values: * $p<0.10$, ** $p<0.05$, *** $p<0.01$.

	Share of firms (%)			Share of pre-treatment emissions (%)		
	Treated	Control	Difference	Treated	Control	Difference
Divestment	22.5	19.5	+3.0	43.5	21.1	+22.4*
Boundary/methodology	27.5	24.0	+3.5	26.4	21.1	+5.3
Acquisitions	0.8	1.0	−0.2	0.0	0.0	+0.0
Output	30.0	32.8	−2.8	40.0	55.3	−15.3
Emission-reduction activities	92.5	92.7	−0.2	98.3	92.7	+5.6
Physical conditions	12.5	10.9	+1.6	13.1	10.1	+3.1
Other/unidentified	29.2	36.2	−7.0	28.1	23.9	+4.2

Reconciling the target and emissions responses. The emissions response is large relative to the target response. At its peak, the estimated effect on the share of firms with a commitment or set target is 7.5 percentage points, an effect equivalent to roughly one in thirteen treated firms committing to or setting a target because of the letter. Yet our estimates imply that the letter lowered aggregate reported emissions by about 18% within four reporting years. Taken at face value, these estimates pair a policy response confined to a limited number of firms with a sizable response in the treatment group's aggregate emissions. Three considerations help to explain this contrast.

First, the emissions response seems to be concentrated among large emitters. As shown above, the aggregate decline appears to be driven largely by a subset of firms that cut reported emissions by half or more (Table 6), so a limited number of responding firms can move aggregate emissions substantially. The reasons firms state in their CDP disclosures are consistent with

this pattern: large treated emitters appear more likely to rely on divestment (Table 7), a route that can lower reported emissions by large amounts within a short time.

Second, the letter might lead firms to reduce emissions without adopting a target. The explicit condition that firms must adopt a science-based target for continued inclusion is a feature of the provider we worked with; other providers do not impose this condition. Climate benchmarks in general reward emission reductions directly: EU rules require climate-transition and Paris-aligned indices to be at least 30% and 50% less carbon-intensive, respectively, than their parent indices. They also require a carbon-intensity reduction of at least 7% annually. Providers meet these requirements through constituent selection and weighting. Firms may therefore have perceived the letter as a broader signal that investors care about their carbon emissions, beyond the specific requirement to adopt a science-based target. The letter may also have encouraged firms that would have adopted a target in any case to follow through with meaningful emission reductions.

Third, firms may reduce emissions while declining to make public pledges. Public sustainability commitments have become politically costlier since 2022 (Garrett and Ivanov, 2023; Gormley, Jha, and Wang, 2024), and firms facing such pressure may have an incentive to decarbonize quietly, a behavior known as *greenhushing*. This is in line with what we observe in our sample: new SBTi commitments fell sharply, from 118 in 2022 to 19 in 2025, and even the index provider we worked with relaxed its strict SBTi-based exclusions in December 2025. A treated firm caught between anti-ESG pressure and climate-screened capital in its investor base may therefore reduce emissions without ever appearing in the SBTi registry.

5 The effect on firm value

We detect no effect of the treatment on firm value in either direction. We form equal-weighted, monthly rebalanced portfolios of the treated and the control firms from monthly returns

and track the long-short (treated-minus-control) portfolio. We report the portfolio's annualized CAPM alpha, the intercept from a regression of the monthly long-short return on the developed-market factor, over expanding windows from the letter, with Newey and West (1987) standard errors using three lags.

Treated and control firms earn indistinguishable stock returns at every horizon over the four years after the letter. Panel (1) of Figure 4 plots the raw cumulative value of equal-weighted portfolios of the two groups, indexed to 100 at the letter date: the two paths track each other closely before and after the letter. Panel (2) reports the long-short portfolio's annualized CAPM alpha over expanding windows, which is statistically indistinguishable from zero at every post-treatment horizon, while the pre-treatment placebo estimates are likewise flat around zero. The point estimates are small, mostly slightly negative, and converge to zero by the fourth year.

The absence of a significant alpha does not depend on the factor model. Table 8 reports the alphas against the CAPM and richer developed-market factor models for February 2022 to January 2026: all are small, no larger than 0.33 percentage points per year in absolute value, and none is statistically significant. The factor loadings are small and insignificant, as randomization equalizes the two arms' factor exposures in expectation. The results are qualitatively similar for value-weighted portfolios: the long-short alphas remain insignificant under every factor model but are substantially less precise (CAPM alpha -0.98 percentage points per year, $p=0.870$, randomization inference; Appendix Figure A6 and Table A30).

The result does not differ between below- and above-median emitters. Table 9 re-estimates the calendar-time alphas separately for firms below and above the median of fiscal-year 2020 emissions, using the same split as in Table 5. Among above-median emitters, where the emissions response is most prominent, the CAPM alpha is 0.44 percentage points per year ($p=0.782$, randomization inference); among below-median emitters it is -1.01 ($p=0.549$, randomization inference). Neither estimate is statistically significant, and the same holds under the other factor models (Appendix Table A31).

Table 8. Calendar-time long-short alphas

Monthly returns of the equal-weighted treated-minus-control portfolio regressed on Ken French developed-market factors, February 2022–January 2026 (48 months): the CAPM (column 1), the Fama–French three-factor model (column 2), the Carhart four-factor model (column 3), and the Fama–French five-factor model (column 4). Alphas are annualized, in percentage points per year; factor loadings are unitless. Newey–West standard errors (3 lags) in parentheses; * $p<0.10$, ** $p<0.05$, *** $p<0.01$. RI p-values are from randomization inference as described in Section 2.2 (2,000 re-assignments of treatment within country blocks), with the long-short portfolio re-formed on each draw.

Factor model	CAPM (1)	FF3 (2)	Carhart (3)	FF5 (4)
Alpha (pp per year)	−0.02	0.32	0.33	0.31
	(0.67)	(0.84)	(0.84)	(0.83)
MKT−RF	0.008	0.002	0.002	−0.003
	(0.010)	(0.014)	(0.014)	(0.018)
SMB		0.015	0.014	0.001
		(0.047)	(0.048)	(0.051)
HML		−0.033	−0.033	−0.026
		(0.031)	(0.032)	(0.046)
UMD			−0.002	
			(0.014)	
RMW				−0.047
				(0.051)
CMA				−0.027
				(0.051)
RI p-value	0.984	0.766	0.758	0.777
Observations (months)	48	48	48	48
R^2	0.006	0.042	0.042	0.072

Table 9. Calendar-time long-short alphas by pre-treatment emissions

Annualized CAPM alphas of the equal-weighted treated-minus-control portfolio (Ken French developed-market factor), February 2022–January 2026 (48 months), for all firms (column 1) and for firms below (column 2) and above (column 3) the median of fiscal-year 2020 carbon emissions in the balanced emissions panel; firms outside that panel enter column (1) only. Alphas are in percentage points per year. Newey–West standard errors (3 lags) in parentheses; * $p<0.10$, ** $p<0.05$, *** $p<0.01$. The randomization-inference p-value is calculated by re-assigning treatment 2,000 times within country blocks among the firms in each column and reconstructing the portfolio on each draw.

	All firms (1)	Below median (2)	Above median (3)
CAPM alpha (pp per year)	−0.02	−1.01	0.44
	(0.67)	(1.12)	(1.58)
RI p-value	0.984	0.549	0.782
Observations (months)	48	48	48
R^2	0.006	0.104	0.050

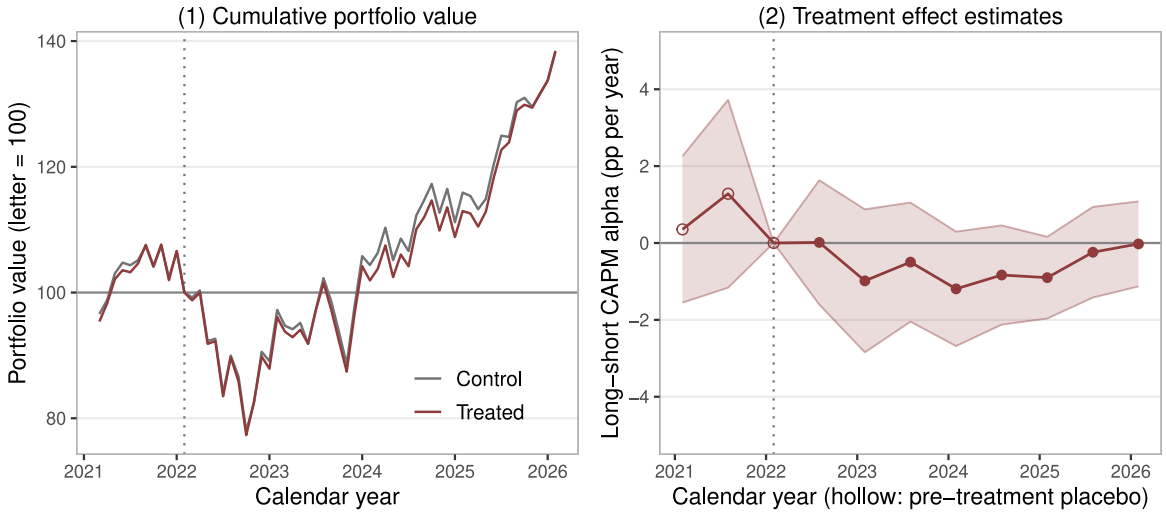


Figure 4. Treatment effect on firm value: portfolio paths and long-short alphas. Panel (1): cumulative value of equal-weighted portfolios of treated and control firms, rebalanced monthly; both portfolios are indexed to 100 at the letter date, and the pre-letter segment is cumulated backward from that point. Panel (2): annualized CAPM alpha of the treated-minus-control portfolio over expanding windows from the letter at half-yearly horizons (filled markers, plotted at the window end) and over placebo windows of twelve and six months ending before the letter (hollow markers, plotted at the window start); the letter-date point is zero by construction. Newey–West standard errors (3 lags); the shaded band is the 90% confidence interval; dotted lines mark the letter date.

The estimates rule out large effects but not small ones. Over the full four-year window, the minimum detectable effect on the annualized long-short alpha is roughly 1.9 percentage points per year. Smaller value effects, plausible for an intervention of this size, would go undetected. The result should therefore be interpreted as the absence of a detectable effect on firm value, not as a precise zero.

6 Discussion

Our experiment demonstrates that index provider engagement influences firm behavior. It is well established that index additions and deletions matter (Shleifer, 1986; Chen, Noronha, and Singal, 2004) and incentivize management to adhere to index rules (Chattopadhyay, Shaffer,

and Wang, 2020). We show that a letter that merely communicates index rules, while holding the rules fixed, influences firms' climate actions. This suggests that index providers can exercise a form of voice through engagement with firms, even though they do not hold shares or voting rights themselves. More broadly, it suggests that index rules are more influential when they are proactively communicated to firms.

Frictions in managerial attention offer one plausible explanation for our results. The index rules were public before the letter, so the letter revealed no information that firms could not have obtained otherwise. That firms nevertheless responded points to a friction between the information available to them and the information they acted on. The friction we find most plausible is scarce attention at the top of the firm: the letter placed the climate-related index rules on the board's agenda. We develop this interpretation below and discuss other frictions consistent with our results.

Attention is a scarce input in economic decisions (Gabaix, 2019; Bordalo, Gennaioli, and Shleifer, 2022). Firms often fail to act on information that is available to them until it is made salient: Hanna, Mullainathan, and Schwartzstein (2014) show that producers change their behavior only after being shown summaries of data they had collected themselves, and Bloom, Eifert, Mahajan, McKenzie, and Roberts (2013) document that firms fail to adopt well-known management practices in part because they are unaware of them. Attention is arguably scarcest at the top of the firm: how chief executives allocate their time predicts firm performance (Bandiera, Prat, Hansen, and Sadun, 2020), and distracted directors weaken oversight (Masulis and Zhang, 2019).

In our setting, a letter from the index provider's CEO to the chair of the board seems well suited to capture scarce attention at the top of the firm: it arrives at the top of the governance hierarchy, addresses the firm specifically, and makes a concrete request with a deadline and consequences, all drawn from an index rule that may previously have received little attention. Our results could thus reflect firms acting on information they had not attended to before at

the board level, rather than on information they did not have. Especially in the context of environmental and social issues, firms face a confusing array of ratings (Berg, Kölbel, and Rigobon, 2022) and vague expectations, making the decision of what to attend to more consequential.

Attention frictions might matter most where the effect of a decision on firm value is unclear ex ante. Many corporate sustainability decisions seem to be of this kind, as the evidence on their value effects is mixed (Krüger, 2015; Flammer, 2015). When decision-makers cannot agree on how likely benefits and costs are, theory predicts that firms underinvest (Garlappi, Giammarino, and Lazrak, 2017); a decision whose effect on firm value is unclear is thus unlikely to become a priority, even if its external effects may be large. Theories of socially responsible investment reach a similar conclusion: investor influence is most effective when the socially responsible action is cheap relative to its social benefit (Broccardo, Hart, and Zingales, 2022; Oehmke and Opp, 2025). Our results are consistent with this view: the letter moved climate commitments and reported emissions, while the decisions it triggered appear, at least ex post, to have mattered little for firm value.

Limited attention is not the only friction that could help to explain our results. As firms also face costs when acquiring and processing information, inattention may be a rational response that leaves decision-makers imperfectly informed (Sims, 2003; Maćkowiak, Matějka, and Wiederholt, 2023). Furthermore, firms turn out to be poorly informed even about matters as fundamental as their macroeconomic environment (Coibion, Gorodnichenko, and Kumar, 2018). Tracking the exclusion rules of many investors and their implications for the firm might be costly; the letter delivered part of this information at no cost to the firm. Agency frictions might offer another explanation. Managers may pursue sustainability out of personal preferences rather than in shareholders' interest (Bénabou and Tirole, 2010; Masulis and Reza, 2015; Cheng, Hong, and Shue, 2023). In our setting, the letter may then have lent external legitimacy to managers who personally favored climate action. These accounts are not mutually exclusive, and our design cannot separate them: the experiment identifies the effect of the

letter, not the friction it alleviated. Nonetheless, the reading that a letter to the board has an effect through making a particular index rule salient seems plausible in light of the evidence.

7 Conclusion

Can index provider engagement on climate-related rules affect corporate climate action? We answer this question with a preregistered field experiment in which the CEO of a major provider of climate benchmarks wrote to the board chairs of 300 constituent firms, randomly selected from 1,227, restating the provider's public policy of excluding firms that lacked a science-based climate target. The letter raises the share of firms that commit to or set such a target by 6.4 percentage points within two years, and treated firms report about 18% lower carbon emissions than control firms four reporting years after the letter, with no differential pre-trend. Firm value shows no detectable response in either direction. Our results suggest that index provider engagement can help to drive corporate climate policy and associated reductions in reported corporate emissions by making existing exclusion rules more salient.

The results suggest that published rules and direct communication are complements: firms responded to a letter that made the exclusion schedule salient to them, and a letter costs little to send. This suggests that it may be worthwhile to communicate index rules proactively, if the goal is that firms adapt policies.

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A Appendix

A.1 Tables

Table A1. Industry composition of the experimental sample and the climate-index universe

Firm counts and shares by NACE sector. The climate-index universe is the full constituent population of the provider's climate indices (1,957 firms); the experimental sample (1,227 firms) excludes Climate Action 100+ firms and firms that had already committed to or set an SBT. The last column is the sample share minus the universe share, in percentage points.

	Climate-index universe		Experimental sample		
NACE sector	N	%	N	%	Diff. (pp)
C - Manufacturing	721	36.8	399	32.5	−4.3
K - Financial and insurance	281	14.4	224	18.3	+3.9
J - Information and communication	231	11.8	153	12.5	+0.7
G - Wholesale and retail trade	150	7.7	95	7.7	+0.1
L - Real estate	130	6.6	86	7.0	+0.4
H - Transport and storage	85	4.3	55	4.5	+0.1
D - Electricity, gas and steam	71	3.6	32	2.6	−1.0
M - Professional and technical	58	3.0	37	3.0	+0.1
B - Mining and quarrying	53	2.7	35	2.9	+0.1
F - Construction	52	2.7	27	2.2	−0.5
N - Administrative and support	31	1.6	23	1.9	+0.3
I - Accommodation and food	30	1.5	10	0.8	−0.7
Q - Human health and social work	24	1.2	22	1.8	+0.6
E - Water, sewerage and waste	15	0.8	9	0.7	−0.0
R - Arts and recreation	14	0.7	10	0.8	+0.1
S - Other services	4	0.2	4	0.3	+0.1
Z - Unclassified	3	0.2	2	0.2	+0.0
A - Agriculture and fishing	2	0.1	2	0.2	+0.1
P - Education	2	0.1	2	0.2	+0.1
Total	1,957	100.0	1,227	100.0	

Table A2. Quarterly treatment effects on corporate climate policies

Treatment coefficients for an SBT commitment or set target (panel 3 of Figure 2) from per-cutoff linear probability models at quarterly cutoffs from April 1, 2022 through January 1, 2026. Columns without controls include treatment only, on all 1,227 firms; columns with controls add the pre-treatment covariates and country and NACE-sector fixed effects and exclude the 16 prior adopters, as in the bottom row of Figure 2. Heteroskedasticity-robust standard errors in parentheses; * $p<0.10$, ** $p<0.05$, *** $p<0.01$. RI p-values are from randomization inference as described in Section 2.2.

	Without controls			With controls		
Cutoff	Estimate	SE	RI p	Estimate	SE	RI p
Apr 2022	−0.010	(0.011)	0.454	−0.005	(0.009)	0.580
Jul 2022	0.003	(0.018)	0.895	0.010	(0.017)	0.519
Oct 2022	0.009	(0.020)	0.745	0.015	(0.020)	0.435
Jan 2023	0.030	(0.023)	0.203	0.041*	(0.023)	0.050
Apr 2023	0.047*	(0.026)	0.060	0.057**	(0.025)	0.018
Jul 2023	0.044	(0.027)	0.089	0.051*	(0.026)	0.048
Oct 2023	0.048*	(0.028)	0.090	0.054**	(0.027)	0.045
Jan 2024	0.064**	(0.029)	0.020	0.074***	(0.028)	0.005
Apr 2024	0.075**	(0.030)	0.007	0.082***	(0.029)	0.004
Jul 2024	0.066**	(0.030)	0.023	0.073**	(0.029)	0.008
Oct 2024	0.067**	(0.031)	0.023	0.071**	(0.029)	0.010
Jan 2025	0.061**	(0.031)	0.038	0.068**	(0.029)	0.016
Apr 2025	0.067**	(0.031)	0.022	0.073**	(0.029)	0.010
Jul 2025	0.068**	(0.031)	0.020	0.074**	(0.029)	0.011
Oct 2025	0.061*	(0.031)	0.045	0.067**	(0.029)	0.022
Jan 2026	0.059*	(0.031)	0.059	0.065**	(0.029)	0.026
Observations		1,227			1,181	

Table A3. Main results without firms deleted after the letter

The three main results on the paper sample and after dropping the 94 firms that Compustat records as deleted after February 1, 2022 (64 through acquisitions or mergers, the rest mostly firms that no longer file), the exclusion foreseen in the preregistration. Panel A: linear probability model for an SBT commitment or set target as of January 1, 2024 with the pre-treatment covariates and country and NACE-sector fixed effects (heteroskedasticity-robust standard errors). Panel B: PPML estimate of equation (2) (standard errors clustered by firm). Panel C: annualized CAPM alpha of the equal-weighted treated-minus-control portfolio, February 2022–January 2026 (Newey–West standard errors, 3 lags). Standard errors in parentheses; * $p<0.10$, ** $p<0.05$, *** $p<0.01$. RI p-values are from randomization inference as described in Section 2.2 (2,000 re-assignments of treatment within country blocks; the portfolio is re-formed on each draw).

	Paper sample	Deleted firms dropped
Panel A: SBT commitment or target as of January 1, 2024 (linear probability model with controls)		
Treatment	0.074***	0.068**
	(0.028)	(0.029)
RI p-value	0.005	0.016
Observations	1,181	1,091
Panel B: Reported carbon emissions, pooled PPML (semi-elasticity)		
Treatment × Post	−0.140**	−0.136**
	(0.061)	(0.062)
RI p-value	0.005	0.007
Firms	1,120	1,081
Panel C: Calendar-time long-short CAPM alpha, 48 months (percentage points per year)		
Alpha	−0.02	0.36
	(0.67)	(0.66)
RI p-value	0.984	0.750
Firms (treated / control)	300 / 927	288 / 845

Table A4. Firms by country and treatment arm

Number and share of sample firms in each country, by treatment arm. Randomization was blocked by country. A χ^2 test of independence of country and assignment yields $X^2 = 3.69$ ($df = 26$, $p = 1$).

	Counts		Frequencies	
Country	Control	Treatment	Control	Treatment
Australia	48	16	0.052	0.053
Austria	6	2	0.006	0.007
Belgium	7	2	0.008	0.007
Brazil	2	0	0.002	0.000
Canada	19	6	0.020	0.020
China	3	2	0.003	0.007
Denmark	3	1	0.003	0.003
Finland	6	2	0.006	0.007
France	23	7	0.025	0.023
Germany	52	17	0.056	0.057
Hong Kong	18	5	0.019	0.017
Ireland	1	1	0.001	0.003
Israel	1	0	0.001	0.000
Italy	22	7	0.024	0.023
Japan	157	51	0.169	0.170
Luxembourg	2	0	0.002	0.000
Netherlands	14	4	0.015	0.013
New Zealand	3	1	0.003	0.003
Norway	7	3	0.008	0.010
Poland	5	2	0.005	0.007
Portugal	2	1	0.002	0.003
Singapore	8	2	0.009	0.007
Spain	9	3	0.010	0.010
Sweden	21	7	0.023	0.023
Switzerland	23	7	0.025	0.023
United Kingdom	39	13	0.042	0.043
United States	426	138	0.460	0.460
Total	927	300	0.999	0.999

Table A5. Firms by sector and treatment arm

Number and share of sample firms in each NACE sector, by treatment arm; firms without a valid NACE code are listed under Z. Sector was not a blocking variable. A χ^2 test of independence of sector and assignment yields $X^2 = 19.33$ ($df = 17$, $p = 0.310$).

	Counts		Frequencies	
Sector	Control	Treatment	Control	Treatment
A	2	0	0.002	0.000
B	23	12	0.025	0.040
C	307	92	0.331	0.307
D	24	8	0.026	0.027
E	6	3	0.006	0.010
F	20	7	0.022	0.023
G	68	27	0.073	0.090
H	45	10	0.049	0.033
I	7	3	0.008	0.010
J	117	36	0.126	0.120
K	169	55	0.182	0.183
L	70	16	0.076	0.053
M	31	6	0.033	0.020
N	12	11	0.013	0.037
P	2	0	0.002	0.000
Q	13	9	0.014	0.030
R	8	2	0.009	0.007
S	3	1	0.003	0.003
Z	0	2	0.000	0.007
Total	927	300	1.000	1.000

Table A6. Post-letter survey respondents and later SBTi outcomes

Treated firms that replied to the index provider's early-2022 outreach (formal survey responses and email replies), compared with all treated firms in the sample ($N = 300$). Outcomes are an SBTi commitment or set target as of January 2023 and January 2025, coded as in the main analysis; firm counts in parentheses.

Group	N	Commit or target, 2023	Commit or target, 2025
Survey respondents	46	20% (9)	39% (18)
Full treated sample	300	15% (46)	33% (99)

Table A7. Treatment effect on corporate climate policies: archived registry downloads

Counterpart of Table 2 with outcomes from archived downloads of the SBTi registry, dated January 31, 2023, January 21, 2024, January 7, 2025, and January 7, 2026. A firm counts as having committed to or set an SBT if any download up to that date lists it with a commitment or a target, including commitments the SBTi later marked as removed. Firms are matched by the identifier-first procedure of the main construction. Odd columns treatment only; even columns with the pre-treatment covariates and country and NACE-sector fixed effects, excluding the 18 firms listed in a download before the letter. Heteroskedasticity-robust standard errors in parentheses; * $p<0.10$, ** $p<0.05$, *** $p<0.01$. RI p-values are from randomization inference as described in Section 2.2.

As of	Jan 2023		Jan 2024		Jan 2025		Jan 2026	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Treatment	0.043	0.054**	0.073**	0.082***	0.062**	0.069**	0.059*	0.067**
	(0.026)	(0.025)	(0.030)	(0.029)	(0.031)	(0.029)	(0.032)	(0.029)
Firm-level controls	No	Yes	No	Yes	No	Yes	No	Yes
Country FE	No	Yes	No	Yes	No	Yes	No	Yes
NACE sector FE	No	Yes	No	Yes	No	Yes	No	Yes
RI p-value	0.098	0.022	0.015	0.003	0.047	0.018	0.056	0.026
Observations	1,227	1,180	1,227	1,180	1,227	1,180	1,227	1,180
R^2	0.002	0.126	0.005	0.156	0.003	0.174	0.003	0.189

Table A8. Treatment effect on reported carbon emissions: no gap filling

Counterpart of Table 5 without filling of interior gaps: the balanced panel comprises the 1,099 firms with reported emissions in every reporting year from 2018 to 2025 before any gap filling. Columns (2) and (3) split firms at the median of fiscal-year 2020 emissions. Coefficients are semi-elasticities. Standard errors clustered by firm in parentheses; * $p<0.10$, ** $p<0.05$, *** $p<0.01$. RI p-values are from randomization inference as described in Section 2.2.

Dependent variable:	Carbon emissions		
	All firms	Below median	Above median
	(1)	(2)	(3)
Treatment × Post	−0.143**	−0.112	−0.143**
	(0.061)	(0.084)	(0.063)
Firm FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
RI p-value	0.005	0.201	0.005
Observations	8,792	4,400	4,392
Within pseudo R^2	0.011	0.003	0.011

Table A9. Treatment effect on reported carbon emissions: pre-treatment window

PPML estimates of equation (2) on three balanced panels that differ in the first included reporting year; column (2) is the panel used throughout the paper. The post period comprises reporting years 2022 onward. Coefficients are semi-elasticities. Standard errors clustered by firm in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. RI p-values are from randomization inference as described in Section 2.2.

Dependent variable:	Carbon emissions		
Reported years:	2017–2025	2018–2025	2019–2025
	(1)	(2)	(3)
Treatment × Post	−0.148**	−0.140**	−0.138**
	(0.063)	(0.061)	(0.060)
Firm FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
RI p-value	0.004	0.005	0.002
Observations	9,801	8,960	7,938
Within pseudo R^2	0.010	0.010	0.011

Table A10. Treatment effect on reported carbon emissions: winsorization rule

PPML estimates of equation (2) under four treatments of the upper tail of emissions: none, and capping at the 99th, 95th, and 90th percentiles within the reporting year; column (3) is the rule used elsewhere in the paper. Coefficients are semi-elasticities. Standard errors clustered by firm in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. RI p-values are from randomization inference as described in Section 2.2.

Dependent variable:	Carbon emissions			
Upper-tail cap:	None	Top 99	Top 95	Top 90
	(1)	(2)	(3)	(4)
Treatment × Post	−0.153*	−0.137*	−0.140**	−0.124***
	(0.080)	(0.075)	(0.061)	(0.047)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
RI p-value	0.080	0.064	0.005	<0.001
Observations	8,960	8,960	8,960	8,960
Within pseudo R^2	0.008	0.007	0.010	0.009

Table A11. Treatment effect on reported carbon emissions, by year: unwinsorized

Counterpart of Table 4 without winsorization of the outcome: estimates of equation (1) with uncapped Scope 1 plus Scope 2 (location-based) emissions; PPML with firm and year fixed effects; coefficients are semi-elasticities. Event time is in reporting years relative to the letter; the omitted reference $t-1$ is reporting year 2021. Standard errors clustered by firm in parentheses; * $p<0.10$, ** $p<0.05$, *** $p<0.01$. RI p-values are from randomization inference as described in Section 2.2.

Dependent variable:	Carbon emissions (unwinsorized)						
Event time:	$t-4$	$t-3$	$t-2$	$t+0$	$t+1$	$t+2$	$t+3$
Reporting year:	2018	2019	2020	2022	2023	2024	2025
Treatment × year	−0.044	−0.023	0.042	−0.049	−0.200**	−0.203**	−0.197*
	(0.113)	(0.101)	(0.065)	(0.057)	(0.095)	(0.096)	(0.110)
RI p-value	0.752	0.864	0.654	0.376	0.012	0.013	0.023
Observations				8,960			
Within pseudo R^2				0.037			

Table A12. Treatment effect on corporate climate policies: interactions with pre-treatment covariates

Linear probability models for an SBT commitment or set target as of January 2024, interacting treatment with an above-median indicator for fiscal-year 2020 carbon emissions (column 1) or emission intensity (column 2); medians over the sample. Both columns include the pre-treatment covariates and country and NACE-sector fixed effects. Heteroskedasticity-robust standard errors in parentheses; * $p<0.10$, ** $p<0.05$, *** $p<0.01$.

Dependent variable:	SBT commitment or set target	
Interaction variable	Carbon emissions	Emission intensity
	(1)	(2)
Treatment	0.050	0.054
	(0.038)	(0.040)
Above-median carbon emissions	0.018	
	(0.034)	
Treatment × Above-median emissions	0.046	
	(0.056)	
Above-median emission intensity		−0.076**
		(0.037)
Treatment × Above-median intensity		0.039
		(0.056)
Firm-level controls	Yes	Yes
Country FE	Yes	Yes
NACE sector FE	Yes	Yes
Observations	1,181	1,181
R^2	0.143	0.145

Table A13. Treatment effect on corporate climate policies by pre-treatment emission intensity

Counterpart of Table 3 splitting firms at the sample median of fiscal-year 2020 emission intensity (carbon emissions per million USD of revenue): linear probability models for an SBT commitment or set target as of January 2024, without and with the pre-treatment covariates and country and NACE-sector fixed effects. Heteroskedasticity-robust standard errors in parentheses; * $p<0.10$, ** $p<0.05$, *** $p<0.01$. RI p-values are from randomization inference as described in Section 2.2.

As of	Jan 2024			
Pre-treatment level	Below median		Above median	
	(1)	(2)	(3)	(4)
Treatment	0.055	0.046	0.077*	0.094**
	(0.042)	(0.040)	(0.041)	(0.041)
Firm-level controls	No	Yes	No	Yes
Country FE	No	Yes	No	Yes
NACE sector FE	No	Yes	No	Yes
RI p-value	0.179	0.252	0.066	0.019
Observations	608	592	607	589
R^2	0.003	0.218	0.006	0.149

Table A14. Treatment effect on corporate climate policies: emissions-weighted

Counterpart of Table 2 weighting each firm by its fiscal-year 2020 carbon emissions; firms without a positive pre-treatment value are excluded. Linear probability models for an SBT commitment or set target at annual cutoffs; odd columns treatment only, even columns with the pre-treatment covariates and country and NACE-sector fixed effects. Heteroskedasticity-robust standard errors in parentheses; * $p<0.10$, ** $p<0.05$, *** $p<0.01$. RI p-values are from randomization inference as described in Section 2.2.

As of	Jan 2023		Jan 2024		Jan 2025		Jan 2026	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Treatment	0.063	0.065	0.142**	0.151***	0.129*	0.130**	0.127*	0.112**
	(0.062)	(0.044)	(0.071)	(0.054)	(0.071)	(0.054)	(0.072)	(0.053)
Firm-level controls	No	Yes	No	Yes	No	Yes	No	Yes
Country FE	No	Yes	No	Yes	No	Yes	No	Yes
NACE sector FE	No	Yes	No	Yes	No	Yes	No	Yes
RI p-value	0.287	0.249	0.033	0.021	0.052	0.039	0.085	0.081
Observations	1,215	1,181	1,215	1,181	1,215	1,181	1,215	1,181
R^2	0.006	0.254	0.021	0.280	0.017	0.290	0.015	0.312

Table A15. BoardEx chair-link spillover: descriptive counts

For treated firms, control firms sharing a board chair (as of February 1, 2022) with at least one treated firm, and all other control firms: number of firms, share of the sample, and share with an SBT commitment or set target as of January 2023. BoardEx data cover 99% of the sample (matched by GVKEY, ISIN, and name).

	Firms	Share of sample	Adoption rate, Jan 2023
Treatment	300	0.244	0.153
Control, chair-linked to treated	11	0.009	0.091
Control, other	916	0.747	0.123

Table A16. BoardEx chair-link spillover: linear probability estimates

Linear probability models for an SBT commitment or set target at annual cutoffs, with the pre-treatment covariates and country and NACE-sector fixed effects. Chair-linked treatment equals one for a control firm sharing a board chair (as of February 1, 2022) with at least one treated firm (GVKEY matched via BoardEx–Compustat). Heteroskedasticity-robust standard errors in parentheses; * $p<0.10$, ** $p<0.05$, *** $p<0.01$. RI p-values are from randomization inference as described in Section 2.2 (2,000 re-assignments of treatment within country blocks), with the chair link re-derived from each permuted assignment, so both rows test the null that the letter moved no firm, addressee or chair-linked peer.

As of	Jan 2023 (1)	Jan 2024 (2)	Jan 2025 (3)	Jan 2026 (4)
Direct treatment	0.040*	0.073***	0.068**	0.065**
	(0.023)	(0.028)	(0.029)	(0.029)
Chair-linked treatment	−0.045	−0.075	0.053	0.030
	(0.101)	(0.121)	(0.127)	(0.127)
Firm-level controls	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes
NACE sector FE	Yes	Yes	Yes	Yes
RI p-value: direct	0.072	0.012	0.022	0.031
RI p-value: chair-linked	0.614	0.493	0.662	0.834
Observations	1,181	1,181	1,181	1,181
R^2	0.089	0.142	0.163	0.183

Table A17. Removal of commitments by the SBTi

Linear probability models for whether a firm's commitment had been removed from the SBTi registry by January 1, 2026, among the 223 firms whose first commitment was at least two years old at that date (the SBTi's 24-month compliance window). Column (1) is treatment only; column (2) adds the pre-treatment covariates and country and NACE-sector fixed effects; column (3) adds commitment year-quarter fixed effects. The sample conditions on committing, itself a treatment outcome. Heteroskedasticity-robust standard errors in parentheses; * $p<0.10$, ** $p<0.05$, *** $p<0.01$.

	(1)	(2)	(3)
Treatment	0.069	0.057	0.060
	(0.061)	(0.064)	(0.065)
Firm-level controls	No	Yes	Yes
Country FE	No	Yes	Yes
NACE sector FE	No	Yes	Yes
Commitment-quarter FE	No	No	Yes
Observations	223	208	208
R^2	0.006	0.284	0.303

Table A18. Treatment effect on corporate climate policies: point-in-time statuses

Counterpart of Table 2 under the point-in-time status coding of Appendix Figure A1: the outcome is a currently listed SBT commitment or target at each January 1, so a commitment removed by the SBTi reverts the firm to no target. Odd columns treatment only; even columns with the pre-treatment covariates and country and NACE-sector fixed effects. Heteroskedasticity-robust standard errors in parentheses; * $p<0.10$, ** $p<0.05$, *** $p<0.01$. RI p-values are from randomization inference as described in Section 2.2.

As of	Jan 2023		Jan 2024		Jan 2025		Jan 2026	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Treatment	0.030	0.041*	0.061**	0.074***	0.052*	0.058**	0.032	0.039
	(0.023)	(0.023)	(0.029)	(0.028)	(0.030)	(0.028)	(0.030)	(0.028)
Firm-level controls	No	Yes	No	Yes	No	Yes	No	Yes
Country FE	No	Yes	No	Yes	No	Yes	No	Yes
NACE sector FE	No	Yes	No	Yes	No	Yes	No	Yes
RI p-value	0.203	0.050	0.034	0.005	0.078	0.034	0.306	0.155
Observations	1,227	1,181	1,227	1,181	1,227	1,181	1,227	1,181
R^2	0.002	0.089	0.004	0.141	0.003	0.172	0.001	0.179

Table A19. Treatment effect on reported carbon emissions: timing of reporting year 2022

PPML estimates of equation (2) with reporting year 2022 (fiscal year 2021, which largely elapsed before the letter) classified as post-treatment (column 1, the classification used in the paper) or as pre-treatment (column 2). Coefficients are semi-elasticities. Standard errors clustered by firm in parentheses; * $p<0.10$, ** $p<0.05$, *** $p<0.01$. RI p-values are from randomization inference as described in Section 2.2.

Dependent variable:	Carbon emissions	
	Post: FY $\geq$ 2021	Post: FY $\geq$ 2022
	(1)	(2)
Treatment × Post	−0.140**	−0.168***
	(0.061)	(0.060)
Firm FE	Yes	Yes
Year FE	Yes	Yes
RI p-value	0.005	0.001
Observations	8,960	8,960
Within pseudo R^2	0.010	0.013

Table A20. Treatment effect on reported carbon emissions: alternative outcomes and estimators

Equation (2) under alternative outcomes and estimators. Column (1) is the main PPML estimate. Columns (2) and (3) estimate the model on the main sample using OLS in levels (tonnes of CO_2e) and in logs; the log specification drops zero-emission firm-years. Column (4) replaces carbon emissions with emission intensity (tonnes of CO_2e per million USD of revenue), estimated by OLS in levels on the intensity-balanced panel. The fit statistic is the within pseudo-R^2 for PPML and the within R^2 for OLS. Standard errors clustered by firm in parentheses; * $p<0.10$, ** $p<0.05$, *** $p<0.01$. RI p-values are from randomization inference as described in Section 2.2.

Dependent variable:	Emission outcomes			
	Carbon emissions			Emission intensity
Estimator:	PPML (main)	OLS, levels	OLS, logs	OLS, levels
	(1)	(2)	(3)	(4)
Treatment × Post	−0.140**	−111,767.6**	−0.136**	−17.63**
	(0.061)	(53,824.3)	(0.058)	(6.909)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
RI p-value	0.005	0.011	0.020	0.002
Observations	8,960	8,960	8,960	8,960
Within (pseudo) R^2	0.010	0.003	0.003	0.005

Table A21. Treatment effect on reported carbon emissions: median-split interaction

PPML estimate of equation (2) augmented with interactions of an above-median indicator for fiscal-year 2020 emissions (the split of Table 5) with the post indicator and with the treatment–post term; the level of the indicator is absorbed by the firm fixed effects. Coefficients are semi-elasticities. Standard errors clustered by firm in parentheses; * $p<0.10$, ** $p<0.05$, *** $p<0.01$. The randomization-inference p-value refers to the triple interaction (2,000 re-assignments of treatment within country blocks).

Dependent variable:	Carbon emissions
	All firms
	(1)
Treatment × Post	−0.102
	(0.084)
Post × Above median	−0.013
	(0.050)
Treatment × Post × Above median	−0.039
	(0.105)
Firm FE	Yes
Year FE	Yes
RI p-value (interaction)	0.690
Observations	8,960
Within pseudo R^2	0.010

Table A22. Treatment effect on reported carbon emissions: by scope

PPML estimates of equation (2) separately for Scope 1 as in the main outcome (Trucost's GHG Direct item, all greenhouse gases), Scope 1 restricted to Kyoto Protocol gases (Trucost's GHG Scope 1 item), Scope 2 (location-based), and Scope 3 (upstream) emissions, each balanced per scope. Coefficients are semi-elasticities. Standard errors clustered by firm in parentheses; * $p<0.10$, ** $p<0.05$, *** $p<0.01$. RI p-values are from randomization inference as described in Section 2.2.

Dependent variable:	Carbon emissions by scope			
	Scope 1 (all GHGs)	Scope 1 (Kyoto GHGs)	Scope 2	Scope 3
	(1)	(2)	(3)	(4)
Treatment × Post	−0.198***	−0.183**	−0.066	0.024
	(0.077)	(0.076)	(0.048)	(0.025)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
RI p-value	0.002	0.004	0.144	0.366
Observations	8,960	8,960	8,960	8,960
Within pseudo R^2	0.016	0.014	0.002	0.001

Table A23. Treatment effect on reported carbon emissions: fully disclosed pre-treatment emissions

PPML estimates of equation (2) for firms with a fiscal-year 2020 Trucost Scope 1 disclosure score of 100. Columns (2) and (3) split these firms at the median of their fiscal-year 2020 emissions. Coefficients are semi-elasticities. Standard errors clustered by firm in parentheses; * $p<0.10$, ** $p<0.05$, *** $p<0.01$. RI p-values are from randomization inference as described in Section 2.2.

Dependent variable:	Carbon emissions		
FY 2020 emissions:	All firms	Below median	Above median
	(1)	(2)	(3)
Treatment × Post	−0.165*	−0.359**	−0.161*
	(0.093)	(0.140)	(0.095)
Firm FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
RI p-value	0.010	0.003	0.023
Observations	3,224	1,616	1,608
Within pseudo R^2	0.020	0.036	0.020

Table A24. Treatment effect on the disclosure share

OLS difference-in-differences with the Trucost Scope 1 disclosure score (0–100) as the outcome, with firm and year fixed effects. The post period comprises reporting years 2022 onward. Standard errors clustered by firm in parentheses; * $p<0.10$, ** $p<0.05$, *** $p<0.01$. RI p-values are from randomization inference as described in Section 2.2.

Dependent variable:	Trucost disclosure score (0–100)
	(1)
Treatment × Post	1.901
	(2.504)
Firm FE	Yes
Year FE	Yes
RI p-value	0.425
Observations	8,936
Within R^2	0.000

Table A25. Treatment effect on reported carbon emissions: firms' CDP-reported emissions

PPML estimates of equation (2) on firms' own CDP-reported carbon emissions, matched to the sample by ISIN. Column (1) uses the balanced panel of firms reporting to CDP in every reporting year; column (2) uses all CDP-matched firm-years. Coefficients are semi-elasticities. Standard errors clustered by firm in parentheses; * $p<0.10$, ** $p<0.05$, *** $p<0.01$. RI p-values are from randomization inference as described in Section 2.2.

Dependent variable:	Carbon emissions	
Sample:	Balanced panel	All CDP-matched firms
	(1)	(2)
Treatment × Post	−0.206*	−0.147*
	(0.110)	(0.086)
Firm FE	Yes	Yes
Year FE	Yes	Yes
RI p-value	0.067	0.021
Observations	1,736	4,159
Within pseudo R^2	0.044	0.018

Table A26. Treatment effect on reported carbon emissions: unbalanced sample

PPML estimates of equation (2) on all 1,226 firms with Trucost data rather than the balanced panel. Columns (2) and (3) split firms at the median of fiscal-year 2020 emissions. Coefficients are semi-elasticities. Standard errors clustered by firm in parentheses; * $p<0.10$, ** $p<0.05$, *** $p<0.01$. RI p-values are from randomization inference as described in Section 2.2.

Dependent variable:	Carbon emissions		
FY 2020 emissions:	All firms	Below median	Above median
	(1)	(2)	(3)
Treatment × Post	−0.134**	−0.153*	−0.133**
	(0.061)	(0.090)	(0.062)
Firm FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
RI p-value	0.008	0.098	0.006
Observations	9,556	4,721	4,811
Within pseudo R^2	0.009	0.004	0.009

Table A27. Treatment effect on reported carbon emissions: attrition bounds

PPML estimates of equation (2) on the 1,179 firms with complete pre-treatment emissions data, imputing the missing post-treatment years of the 59 firms that stop reporting by forward-filling each firm's last observed level along an extreme same-arm trajectory. Median trajectory: both arms at their own arm's median post-period emissions growth. Adversarial: imputed control firms follow the 5th percentile of post-period growth among control firms with complete data, imputed treated firms the treated-arm 95th percentile. Favorable: the reverse. The post period comprises reporting years 2022 onward. Coefficients are semi-elasticities. Standard errors clustered by firm in parentheses; * $p<0.10$, ** $p<0.05$, *** $p<0.01$. RI p-values are from randomization inference as described in Section 2.2 (2,000 re-assignments of treatment within country blocks); the arm-specific trajectories and the imputation are recomputed on each draw.

Dependent variable:	Carbon emissions		
	Median trajectory	Adversarial	Favorable
	(1)	(2)	(3)
Treatment × Post	−0.135**	−0.131**	−0.139**
	(0.061)	(0.061)	(0.061)
Firm FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
RI p-value	0.003	0.004	0.003
Observations	9,432	9,432	9,432
Within pseudo R^2	0.009	0.008	0.009

Table A28. Leave-one-firm-out estimates: pooled specification

Distribution of the Treatment × Post coefficient of equation (2) across 1,120 re-estimations, each dropping one firm. p-values are based on standard errors clustered by firm.

Statistic	Value
Treatment × Post (full sample)	−0.140
Standard error (full sample)	0.061
p-value (full sample)	0.022
Number of LOO regressions	1,120
LOO minimum	−0.155
LOO median	−0.140
LOO maximum	−0.112
Share of LOO regressions with $p<0.05$	100.0%
Share of LOO regressions with $p<0.10$	100.0%

Table A29. Leave-one-firm-out estimates: year-specific specification

Distribution of the t+3 coefficient of equation (1) across 1,120 re-estimations, each dropping one firm. p-values are based on standard errors clustered by firm.

Statistic	Value
Treatment ×t+3 (full sample)	−0.192
Standard error (full sample)	0.072
p-value (full sample)	0.007
Number of LOO regressions	1,120
LOO minimum	−0.205
LOO median	−0.192
LOO maximum	−0.165
Share of LOO regressions with $p<0.05$	100.0%
Share of LOO regressions with $p<0.10$	100.0%

Table A30. Calendar-time long-short alphas: value-weighted portfolios

Counterpart of Table 8 with value-weighted portfolios: each month, firms are weighted by their market capitalization at the end of the previous month (firms without one are excluded for that month). Monthly returns of the treated-minus-control portfolio regressed on Ken French developed-market factors, February 2022–January 2026 (48 months). Alphas are annualized, in percentage points per year; factor loadings are unitless. Newey–West standard errors (3 lags) in parentheses; * $p<0.10$, ** $p<0.05$, *** $p<0.01$. RI p-values are from randomization inference as described in Section 2.2 (2,000 re-assignments of treatment within country blocks), with the long-short portfolio re-formed on each draw.

Factor model	CAPM	FF3	Carhart	FF5
	(1)	(2)	(3)	(4)
Alpha (pp per year)	−0.98	2.32	2.60	2.35
	(2.56)	(2.54)	(2.74)	(2.41)
MKT−RF	−0.102***	−0.149***	−0.162***	−0.157***
	(0.032)	(0.026)	(0.032)	(0.030)
SMB		0.270**	0.215	0.243**
		(0.104)	(0.137)	(0.098)
HML		−0.260***	−0.243**	−0.267
		(0.090)	(0.109)	(0.166)
UMD			−0.094	
			(0.120)	
RMW				−0.108
				(0.182)
CMA				−0.020
				(0.152)
RI p-value	0.870	0.741	0.706	0.752
Observations (months)	48	48	48	48
R^2	0.060	0.196	0.211	0.203

Table A31. Calendar-time long-short alphas by pre-treatment emissions: alternative factor models

Counterpart of Table 9 under the CAPM, the Fama–French three-factor model, the Carhart four-factor model, and the Fama–French five-factor model (Ken French developed-market factors): annualized alphas of the equal-weighted treated-minus-control portfolio, February 2022–January 2026 (48 months), for all firms and for firms below and above the median of fiscal-year 2020 carbon emissions in the balanced emissions panel. Alphas are in percentage points per year. Newey–West standard errors (3 lags) in parentheses; * $p<0.10$, ** $p<0.05$, *** $p<0.01$. The randomization-inference p-value is calculated by re-assigning treatment 2,000 times within country blocks among the column's firms and reconstructing the portfolio on each draw.

	All firms	Below median	Above median
	(1)	(2)	(3)
CAPM alpha (pp per year)	−0.02	−1.01	0.44
	(0.67)	(1.12)	(1.58)
RI p-value	0.984	0.549	0.782
FF3 alpha (pp per year)	0.32	−0.90	1.02
	(0.84)	(1.45)	(1.71)
RI p-value	0.766	0.581	0.485
Carhart alpha (pp per year)	0.33	−0.89	1.05
	(0.84)	(1.51)	(1.76)
RI p-value	0.758	0.577	0.460
FF5 alpha (pp per year)	0.31	−0.85	0.99
	(0.83)	(1.59)	(1.81)
RI p-value	0.777	0.609	0.503
Observations (months)	48	48	48

A.2 Figures

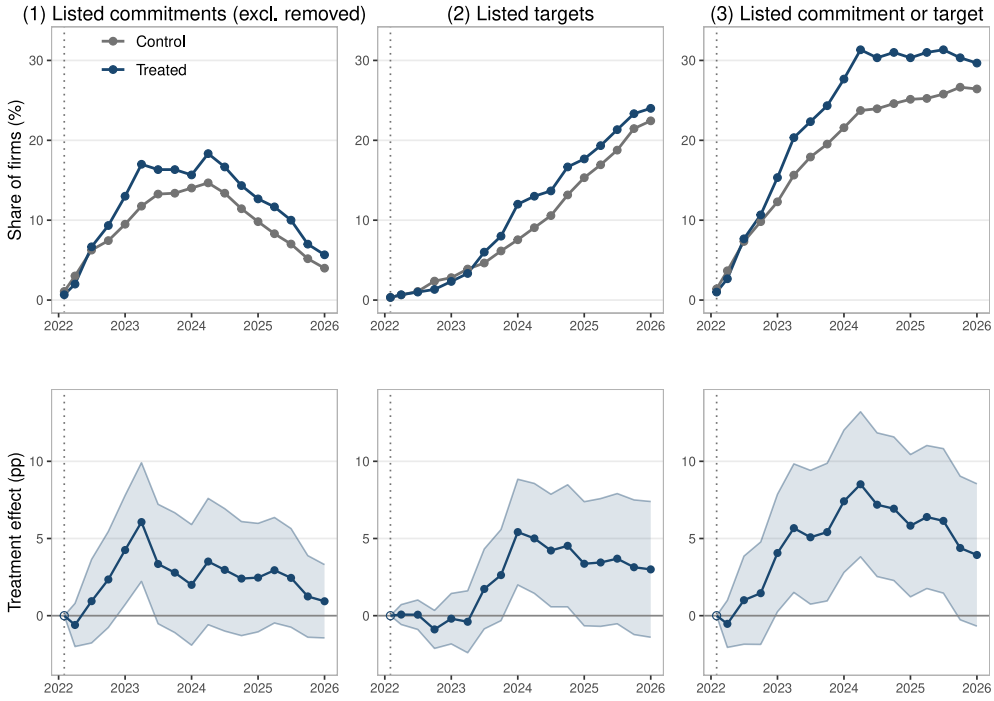


Figure A1. Dynamics of the commitment response: point-in-time statuses. Counterpart of Figure 2 under point-in-time status coding, in which a commitment removed by the SBTi reverts the firm to no target: shares of firms by arm with a currently listed commitment (panel 1), a currently listed target (panel 2), and either (panel 3) at the letter date and quarterly cutoffs from April 2022 through January 2026 (top row), and per-cutoff linear-probability treatment effects with the full control set (bottom row). Panel 2 is identical under both codings. Construction as in Figure 2; shaded bands are 90% heteroskedasticity-robust confidence intervals. Appendix Table A18 tabulates the corresponding estimates.

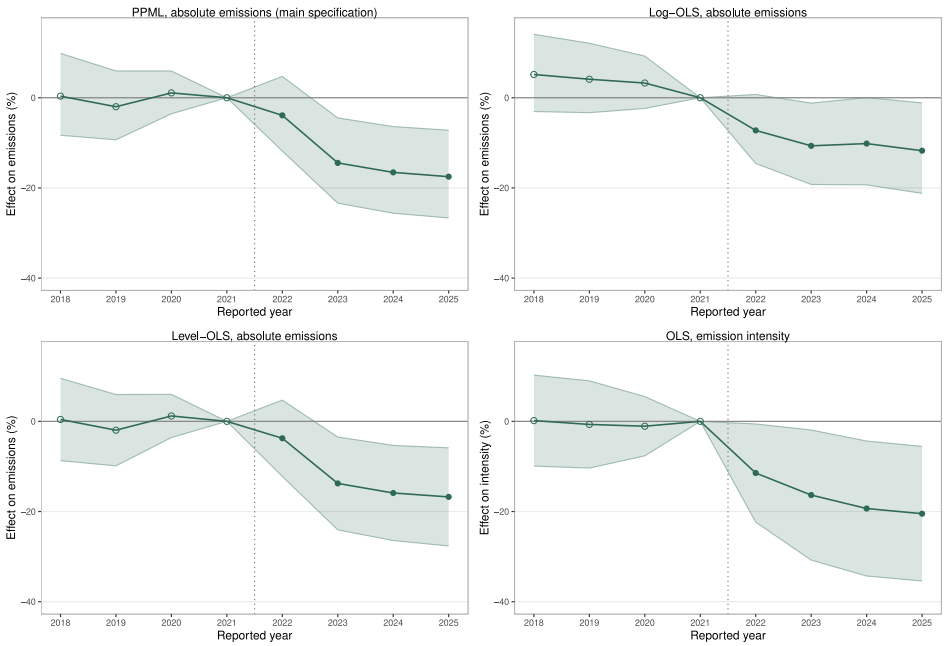


Figure A2. Event studies under alternative estimators: all firms. Year-specific difference-in-differences effects on carbon emissions under four estimators: the main PPML specification; OLS in levels and in logs (Table A20, columns 2 and 3); and OLS in emission-intensity levels on the intensity sample (Table A20, column 4). Effects are implied percent changes: $100(e^{\beta}-1)$ for PPML and log OLS, and 100β over the pre-treatment control mean for the two levels estimators. Reporting year 2021 is the reference; the horizontal axis is the reporting year and the dotted line marks the letter date. Shaded areas are 90% confidence bands from standard errors clustered by firm. Joint Wald tests of the pre-treatment coefficients yield $p=0.64$ (PPML), 0.76 (log OLS), 0.64 (level OLS), and 0.97 (intensity). The four panels share one vertical scale, which differs across Figures A2–A4.

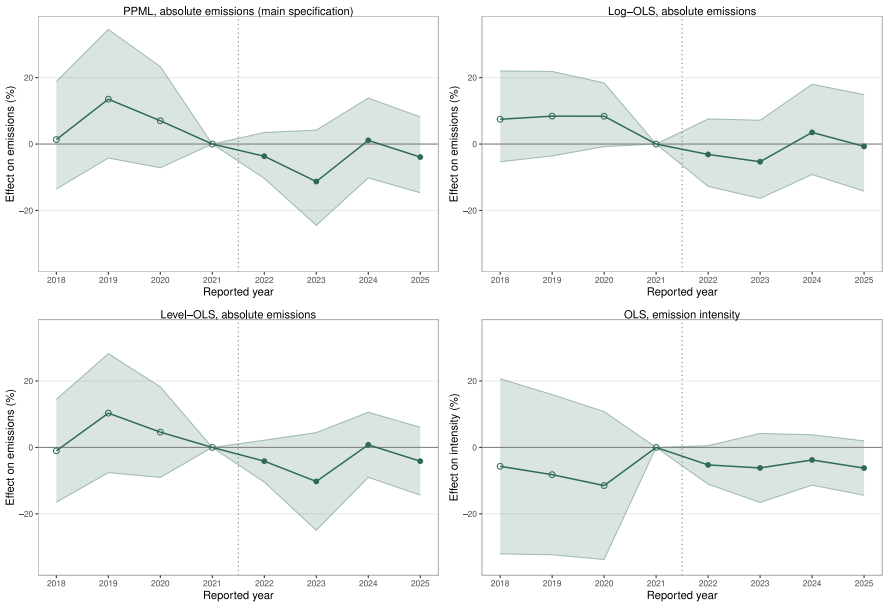


Figure A3. Event studies under alternative estimators: below-median emitters. As Figure A2, for firms below the median of the fiscal-year 2020 outcome (carbon emissions for the three emissions estimators; emission intensity for the intensity estimator). Joint Wald tests of the pre-treatment coefficients yield $p=0.33$ (PPML), 0.51 (log OLS), 0.44 (level OLS), and 0.79 (intensity). The four panels share one vertical scale.

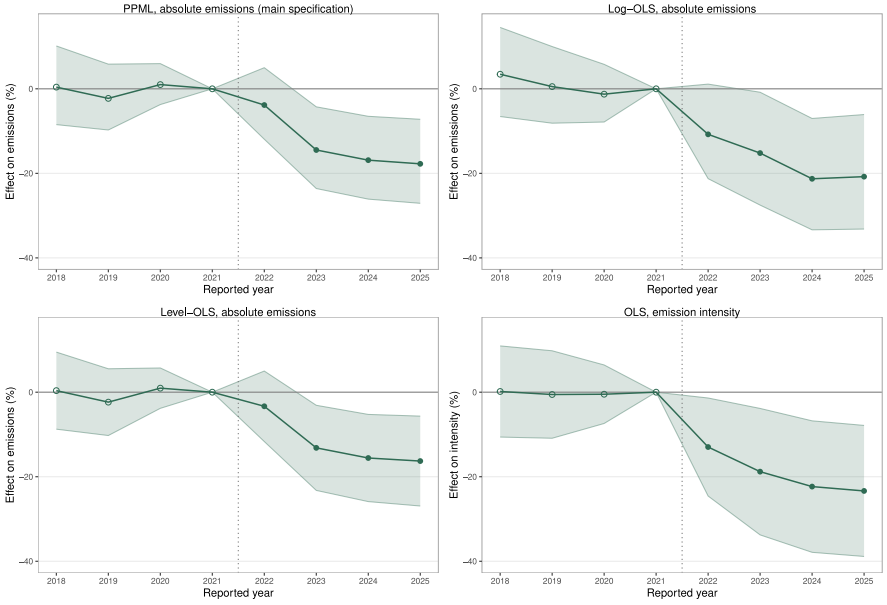


Figure A4. Event studies under alternative estimators: above-median emitters. As Figure A2, for firms above the median of the fiscal-year 2020 outcome (carbon emissions for the three emissions estimators; emission intensity for the intensity estimator). Joint Wald tests of the pre-treatment coefficients yield $p=0.59$ (PPML), 0.73 (log OLS), 0.59 (level OLS), and 0.99 (intensity). The four panels share one vertical scale.

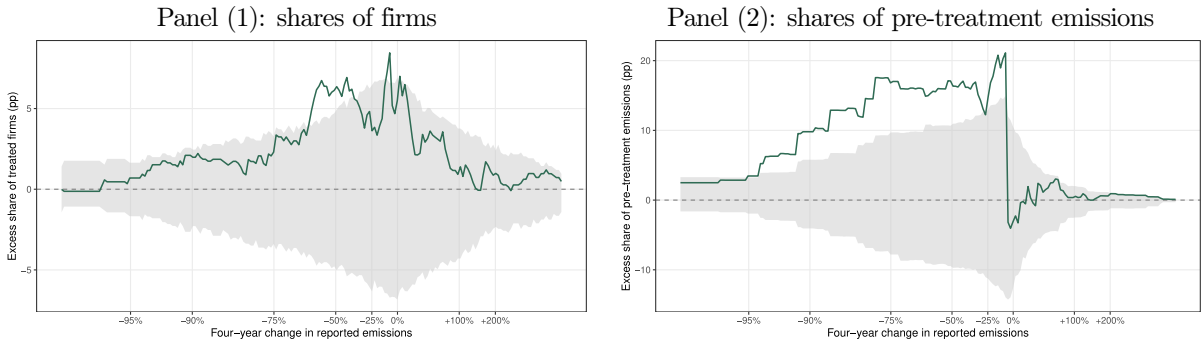


Figure A5. Difference in the cumulative distributions of emission changes. For each level of the four-year change in reported emissions (reporting years 2021 to 2025; log scale, labeled in percent; negative values are reductions), the treated-minus-control difference in the share of the arm's firms (panel 1) and of the arm's pre-treatment emissions (panel 2) with a change at most that large, in percentage points; on the reduction side, positive values mean that a larger treated than control share reaches at least that reduction. Firms with positive pre-treatment emissions (1,120). Shaded areas are pointwise 95% randomization null bands from 2,000 re-assignments of treatment within country blocks; Kolmogorov–Smirnov-type joint tests yield $p=0.062$ (panel 1) and $p=0.009$ (panel 2).

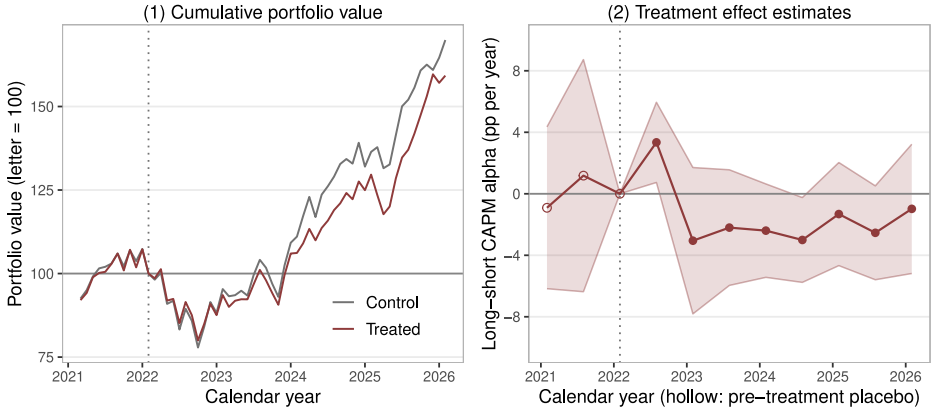


Figure A6. Firm value: value-weighted portfolios. Counterpart of Figure 4 with value-weighted portfolios: each month, firms are weighted by their market capitalization at the end of the previous month. Panel (1): cumulative value of the treated and control portfolios, rebalanced monthly, indexed to 100 at the letter date, with the pre-letter segment cumulated backward from that point. Panel (2): annualized CAPM alpha of the treated-minus-control portfolio over expanding windows from the letter date at half-yearly horizons (filled markers, plotted at the window end) and over placebo windows of twelve and six months ending before the letter (hollow markers, plotted at the window start); the letter-date point is zero by construction. Newey–West standard errors (3 lags); the shaded band is the 90% confidence interval; dotted lines mark the letter date.

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